



USER MANUAL · VERSION 0.6

# EZ2DEMOSCENE

Build seamless, stylised 3D loops the demoscene way: neon, mirrors, particles and retro pixels. No code needed.

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For Windows, Linux, macOS, the web & Android · Covers the editor, the node graph, exporting and the command line

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# 1 Welcome

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EZ2DEMO SCENE is a tool for making **short, perfectly looping 3D animations** in the style of the demoscene: glowing neon arenas, mirrored golden halls, glossy crystals orbited by debris, raymarched tunnels, kaleidoscopes and synthwave sunsets. The loops are for music visuals, VJ sets, wallpapers, social posts and GIFs.

You never write code or shaders. You pick a starting scene, then change it with sliders, colour pickers and menus. When you like it, you export an MP4, WebM, GIF or a folder of PNG frames.



## Neon Arena

Black gloss, red neon, crystal skyline.



## Gold Kaleido Room

Mirrored hall, orb walls, chandelier.



## Orbiting Solid

Blue dodecahedron and a debris swarm.



## Retro Tunnel

XOR tunnel, EGA palette, CRT.



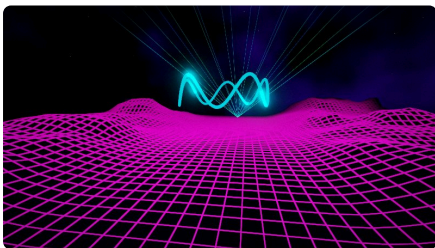
## Plasma Kaleidoscope

Oldschool plasma, spinning tori.



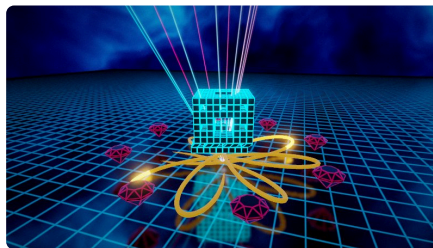
## Synth Sunset

Striped sun, neon grid, pyramids.



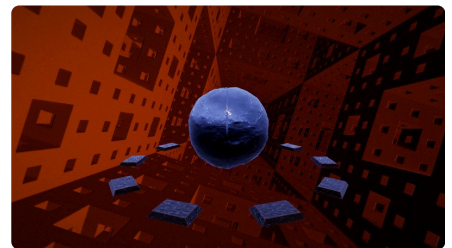
## Vector Valley

Wireframe landscape, laser fans, neon ribbon.



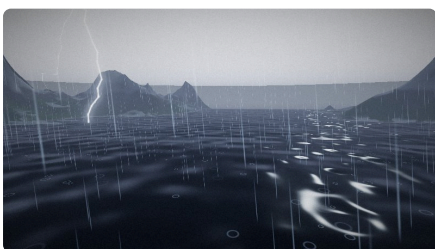
## Glitch Shrine

Glitching sponge, flashing gems, pulsing rose.



## Sponge Dive

Menger sponge flight, breathing displaced orb.



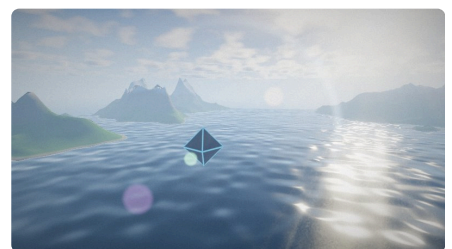
## Stormy Lake

Rain, lightning and storm clouds over a lake.



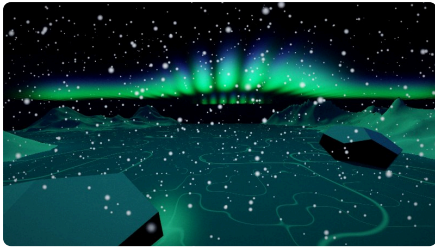
## Lava World

Lava rivers in volcanic canyons, embers.



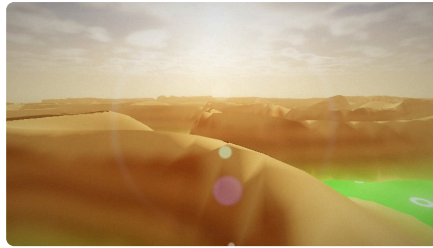
## Sunbeam Peaks

Volumetric clouds, god rays, lens flare.



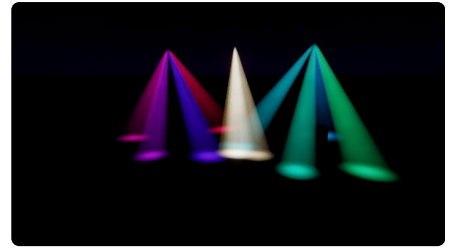
### Aurora Tundra

Northern lights, frozen lake, snow.



### Dune Sea

Sand dunes, sandstorm, a toxic oasis.



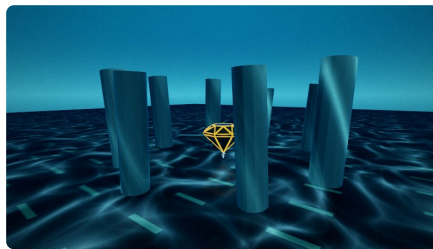
### Club Spotlights

Spotlight cones, light pools, mist.



### Rainbow Falls

Day and night, a rainbow, a waterfall.



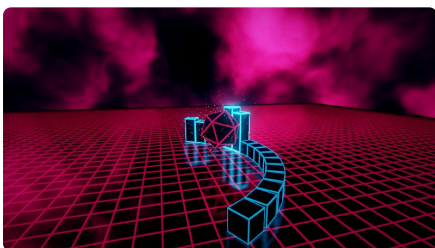
### Sunken Temple

Underwater caustics and bubbles.



### Twister

A tornado over wet, stormy plains.



### Music Reactor

Load a song: equalizer, kick flashes, time warp.

## The one big idea: everything loops

A scene is exactly **a whole number of beats** long, at a tempo you choose (BPM). Every motion in EZ2DEMO SCENE completes a whole number of times during that loop. That includes spins, orbits, pulses, particles, scrolling textures, drifting clouds and even film grain. So the last frame leads straight back into the first, with no jump.

### Why this matters

You never have to "fix the loop" by hand. Whatever you change, the export repeats seamlessly. This is why many settings ask for a count of *turns per loop* or *cycles per loop* rather than a speed.

## How a scene is built

A scene is a stack of **layers** drawn together:

- **Background:** the sky or environment behind everything (nebula, tunnel, plasma...).
- **Mirror floor:** a reflective ground plane, optionally with a glowing grid.
- **Shapes:** 3D objects, possibly copied many times in rings, grids, walls or swarms.

- **Particles:** sparks, stars, glitter, vortices.

On top of the layers sit the **camera**, the **light & fog**, and the **post effects** that give the final look (glow, kaleidoscope, retro palettes, CRT...).

## 2 Installing & launching

### No install needed

You can also use EZ2DEMOSCENE in a web browser or on an Android phone. See chapter 15.

### What you need

| Item                                             | Details                                                                                                                                                                                 |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A graphics card                                  | Any GPU with Vulkan, DirectX 12 or Metal support (most GPUs from 2015 onward). A software renderer also works, only slower.                                                             |
| Rust 1.95 or newer ( <i>source builds only</i> ) | Needed to build the app from source. Install it from <code>rustup.rs</code> , or update with <code>rustup update stable</code> .                                                        |
| ffmpeg ( <i>optional</i> )                       | Needed only for MP4, WebM and GIF export. Download it from <code>ffmpeg.org</code> and make sure the <code>ffmpeg</code> command works in a terminal. PNG sequences need nothing extra. |
| Linux only                                       | <code>sudo apt install libasound2-dev libxkbcommon-dev libwayland-dev</code>                                                                                                            |

### The easy way: download a release

Ready-made downloads are published on the project's GitHub **Releases** page for Windows, Linux and macOS. Unzip the archive and run `ez2demoscene` ( `ez2demoscene.exe` on Windows). The Windows and Linux downloads include **ffmpeg**, so video and GIF export work straight away. On macOS, install it with `brew install ffmpeg`. The archive also contains this manual, the presets and the texture pack.

### Building from source

```
git clone https://github.com/Roganis/EZ2DEMOSCENE
cd EZ2DEMOSCENE
cargo run --release          # opens the editor
cargo run --release -- my.ez2.json  # opens a saved project
```

The first build downloads and compiles libraries, so it takes a few minutes. Later starts are instant. The finished program is `target/release/ez2demoscene` ( `ez2demoscene.exe` on Windows), and you can copy it anywhere.

When the editor opens you are greeted by the **preset gallery**.



The preset gallery appears at start-up. Hover a card for a description, click to load it.

### 3 A tour of the window



The editor with the Neon Arena preset and the Camera selected.

| Area               | What it does                                                                                                                                                                                                                                                                                                                                                             |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Top bar            | <b>File</b> , <b>Edit</b> , <b>Presets</b> , 🎲 <b>Randomize</b> (with its ⚙ settings), <b>Surprise me</b> , the <b>Simple</b> / <b>Nodes</b> mode switch, ● <b>Export</b> and ? help.                                                                                                                                                                                    |
| Scene panel (left) | The project <b>name</b> ; the scene-wide sections ( <b>Timing &amp; music</b> , <b>Camera</b> , <b>Light &amp; fog</b> , <b>Post effects</b> , <b>Your images</b> ); and the <b>layer list</b> with + <b>Add</b> .                                                                                                                                                       |
| Viewport (centre)  | A live preview. <b>Drag</b> to turn the camera and <b>scroll</b> to zoom. Above it you can pick the <b>aspect ratio</b> (16:9, 4:3, 1:1, 9:16, 21:9), the <b>preview quality</b> , a live <b>fps</b> readout, the <b>Move</b> / <b>Rotate</b> / <b>Scale</b> tools and the <b>Grid</b> switch (see 5.6). Click an object to select it. Status messages also appear here. |
| Inspector (right)  | All the settings of whatever is selected on the left.                                                                                                                                                                                                                                                                                                                    |
| Timeline (bottom)  | Play/pause, back to start, a scrubber with <b>beat ticks</b> (tall ticks mark every 4 beats), the current beat, BPM and loop length.                                                                                                                                                                                                                                     |

#### Tip

Hover almost any label in the inspector to see a short explanation.

## 4 Quick start: your first loop in five minutes

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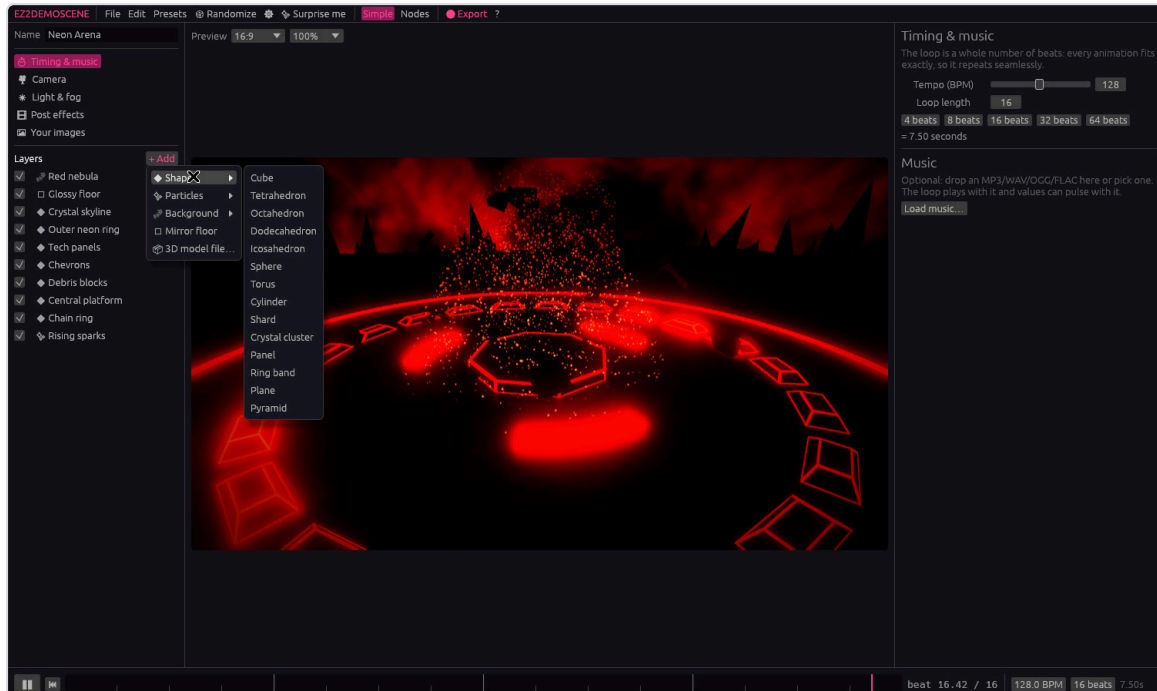
- 1 **Pick a preset.** In the gallery, click **Orbiting Solid**. The scene starts playing straight away.
- 2 **Change a colour.** Click the **Dodecahedron** layer on the left. In the inspector under **Material**, click the **Colour** swatch and pick a hot pink.
- 3 **Make it glow on the beat.** Still in **Material**, find **Glow** and click the ~ button next to it. Set the wave to **Pulse**, **amount** to 3 and **x/loop** to **8** (the loop is 8 beats long). It now flashes on every beat.
- 4 **Add particles.** Click + **Add** → ✨ **Particles** → **Vortex**. In the new layer, set **Colour (young)** to white and **Colour (old)** to pink.
- 5 **Add a retro touch.** Select **Post effects** on the left and tick **Retro palette**. Choose **PICO-8**.
- 6 **Export.** Press **Ctrl + E**. Pick **Animated GIF** (the size switches to 480×270, a good size for GIFs), choose a file with **Browse...** and press • **Export**.
- 7 **Save your project** with **Ctrl + S** to keep editing it later.

### Made a mess?

**Ctrl + Z** undoes any change, including randomizing and loading presets.

# 5 Layers

The layer list works like the layers in a paint program. Each row has a **visibility** tick box and the layer's name. Click a row to edit it. Under the list, the buttons     move, duplicate or delete the selected layer, and **right-clicking** a layer shows the same options.



+ **Add** offers shapes, particles, backgrounds, terrain, lasers, ribbons, weather, a mirror floor and 3D model files.

## 5.1 Backgrounds

A background fills the whole screen behind the other layers and follows the camera. Every background has three colours (**Colour A** is usually the darkest), a **Motion / loop** count, a **Brightness** and a **Detail** (pattern scale).

| Style                | Description                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Gradient sky         | A soft two-colour sky with a horizon glow and a sun halo in the light direction.                                                   |
| Nebula clouds        | Swirling fractal clouds with stars. The clouds drift on a closed path so they loop.                                                |
| Starfield            | Twinkling stars over faint coloured dust.                                                                                          |
| Raymarched tunnel    | The classic demo tunnel flying forward through a winding tube. Pick a <b>Wall texture</b> (the XOR or checker textures are great). |
| Raymarched fractal   | A volumetric "Kaliset" fractal space. It looks psychedelic and cosmic, and is heavy on slow GPUs.                                  |
| Oldschool plasma     | The 90s sine plasma.                                                                                                               |
| Synthwave sun & grid | A striped sun sinking into a scrolling neon grid, with stars.                                                                      |

|                          |                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Raymarched sponge flight | Flying through an endless Menger sponge, a lattice of beams or a field of cubes. Heavy on phones: lower <b>Quality</b> if it stutters.                                                                                                                                                                                           |
| Raymarched ring corridor | Flying through glowing rings, squares or triangles that alternate between colours B and C.                                                                                                                                                                                                                                       |
| Volumetric clouds        | A real sky: a gradient from colour B (horizon) to A (overhead), a sun in the light direction and raymarched 3D clouds lit by it, glowing at the edges when you look towards the sun. Colour C tints the shadowed sides. The clouds drift with <b>Motion / loop</b> . Pair it with <b>God rays</b> for sunbeams. Heavy on phones. |
| Aurora night sky         | Stars and rippling northern-lights curtains that fade from colour B at the bottom to colour C at the top. Colour A is the night sky.                                                                                                                                                                                             |

### Tip

You can stack two backgrounds, but the later one covers the earlier one (and only that one is drawn). Usually one is enough.

**Resolution** renders a background at half or quarter resolution and upscales it. For volumetric clouds and the raymarched styles this is 4× or 16× less work and hardly visible; new clouds, fractal and sponge backgrounds start at half.

## Raymarching settings

The four raymarched backgrounds (tunnel, fractal, sponge, rings) get an extra **Raymarching** section. Only the settings that matter for the chosen style are shown, and all of them except Variant, Wall pattern, Roll and Quality can be animated with ~.

| Setting     | Tunnel                                                                         | Fractal                       | Sponge                                  | Rings                         |
|-------------|--------------------------------------------------------------------------------|-------------------------------|-----------------------------------------|-------------------------------|
| Variant     | Round, Square, Hexagon, Triangle, Flower                                       | Kaliset, Crystal, Nebula      | Menger sponge, Beam lattice, Cube field | Rings, Squares, Triangles     |
| Size        | Radius                                                                         | Zoom                          | Cell size                               | Ring size                     |
| Twist       | Twists the walls along the flight                                              | n/a                           | Twists the space                        | Turns each ring a little more |
| Warp        | Wall wobble                                                                    | Fold (changes the whole look) | n/a                                     | Thickness                     |
| Bend        | How much the path winds                                                        | Drift                         | Path sway                               | Path sway                     |
| Glow        | Light rings sliding past                                                       | Brightness                    | Edge glow                               | Glow                          |
| Fog ×       | Depth fog (0 = none).                                                          |                               |                                         |                               |
| Roll / loop | Whole turns of the view per loop.                                              |                               |                                         |                               |
| Quality     | Raymarch steps (fractal: iterations). 0 uses the default; lower it for phones. |                               |                                         |                               |

The tunnel also has a **Wall pattern** (XOR, Checker, Rings, Stripes) used when there is no wall texture.

**Sky settings**

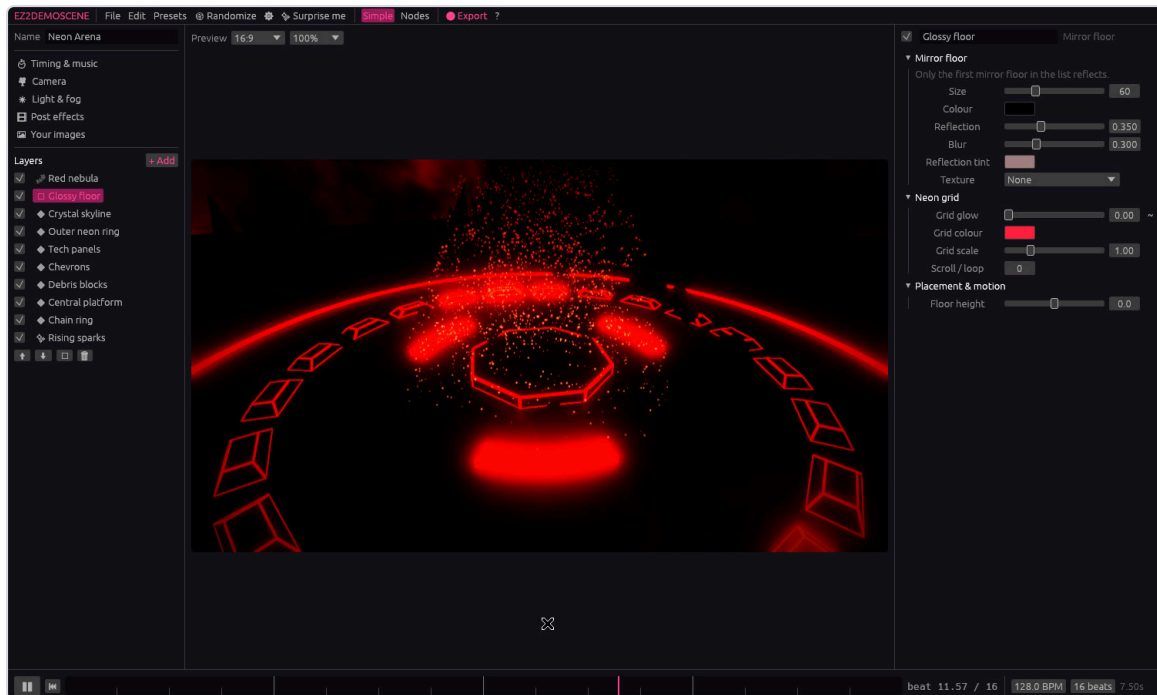
The two skies share the same section, with their own meanings:

| Setting | Volumetric clouds                                   | Aurora night sky                                |
|---------|-----------------------------------------------------|-------------------------------------------------|
| Variant | Fluffy, Overcast, Storm (dark and heavy)            | Curtains, Rings (around the zenith), Spiral     |
| Size    | Cloud scale                                         | Height of the curtains                          |
| Twist   | n/a                                                 | Sway: how much the curtains swing over the loop |
| Warp    | Coverage: how much of the sky is cloudy             | Ripples: the fine vertical rays                 |
| Bend    | Thickness of the cloud layer                        | n/a                                             |
| Glow    | Sun glow around the sun and through the clouds      | Brightness                                      |
| Fog x   | Haze: how much distant clouds melt into the horizon | n/a                                             |
| Quality | Raymarch steps (default 28)                         | n/a                                             |

**Loops**

Clouds can drift any distance and still loop: two copies of the cloud field, half a loop apart, cross-fade into each other, so the jump back at the end of the loop happens while it is invisible.

## 5.2 Mirror floor



Mirror floor settings.

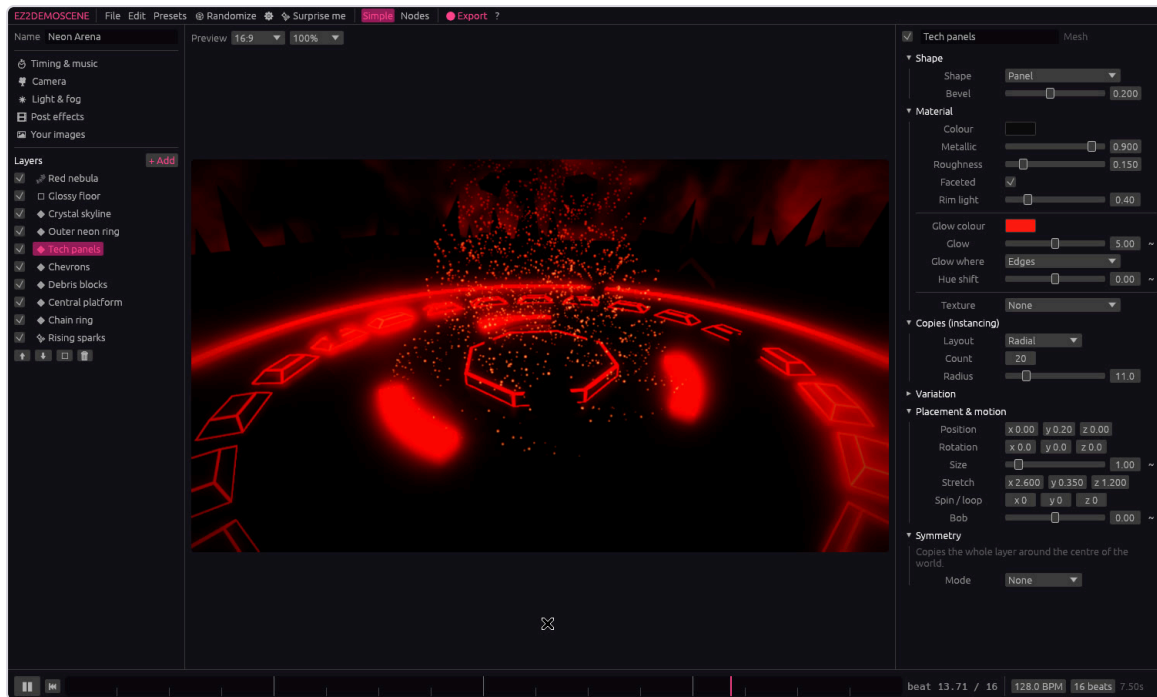
A large reflective ground plane. The scene is really reflected in it, including particles and the sky.

- **Reflection**: 0 is matte, 1 is a perfect mirror. **Blur** gives a frosted, polished-stone look. **Reflection tint** colours the reflection (gold for a gold room).
- **Texture** and **Tiling** put a pattern on the floor. The `led_grid` texture makes an LED dance floor.
- **Neon grid** draws glowing lines. Set **Grid glow**, **Grid colour**, **Grid scale** and **Scroll / loop** (cells per loop) for the synthwave "driving" effect.
- **Floor height** (in **Placement & motion**) moves the floor up or down.

### One mirror

Only the **first** mirror floor in the list reflects. Extra ones are ignored.

## 5.3 Shapes



A shape layer (Tech panels): Shape, Material, Copies, Variation, Placement & motion, Symmetry.

In **Shape** choose one of 25 built-in shapes, or a 3D file:

| Shape                                                    | Good for                                                                            |
|----------------------------------------------------------|-------------------------------------------------------------------------------------|
| Cube, Tetrahedron, Octahedron, Dodecahedron, Icosahedron | Classic demo solids. Try them with <b>Faceted</b> on and <b>Glow where: Edges</b> . |
| Sphere (Detail 0-5)                                      | Orbs and bubbles. Detail 0 gives a faceted low-poly ball.                           |
| Torus                                                    | Rings and chain links (Thickness, Segments).                                        |
| Cylinder (Sides)                                         | Platforms and pillars. Eight sides makes an octagonal stage.                        |
| Shard, Crystal cluster                                   | Sharp spikes and crystal mountains. <b>Seed</b> gives a different random shape.     |
| Panel (Bevel)                                            | Sci-fi plates, light panels, screens.                                               |
| Ring band (Arc, Width, Height)                           | Neon rings and arcs. Set Arc below 360° for partial arcs and chevrons.              |
| Plane, Pyramid, Cone                                     | Simple building blocks.                                                             |
| Capsule (Length), Rounded cube (Roundness)               | Soft, toy-like objects and pills.                                                   |
| Torus knot (Loops, Twists), Spring (Turns), Möbius strip | Twisty tubes and bands. Knots with more loops and twists get more intricate.        |
| Star (Points, Inner radius), Gear (Teeth), Heart         | Flat extruded logos and badges ( <b>Depth</b> sets the thickness).                  |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gem (Facets)              | A brilliant-cut diamond. Lovely with <b>Glow where: Edges</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Menger sponge (Level 0–3) | The classic fractal cube. Level 3 has 8,000 cubes, so use it sparingly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3D text                   | Solid letters for logos: type the text (several lines are fine), pick a font (Pixel gives chunky voxel letters) or a TTF/OTF file, and set the <b>Depth</b> . They take every material, relief, glitch, deform and copies option.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Raymarched shapes         | <b>Metaballs</b> (balls melting together as they orbit), <b>Gyroid</b> (a ball carved into a flowing lattice), <b>Fractal bulb</b> (the Mandelbulb) and <b>Melting box</b> (two balls passing through a rounded box). They are worked out per pixel, so they are perfectly smooth, cut into other shapes, floors and terrain correctly, cast and take sun shadows and use the layer's colour, metal, roughness, rim, glow and colour ramp (with <b>Glow where</b> other than Whole, the creases glow). <b>Motion</b> sets how many times their own movement repeats per loop (0 = still), and copies each get their own variation. Textures, relief, deform and glitch don't apply. They cost per pixel they cover, so a few big ones or a dozen small ones are fine; the <b>Liquid Metal</b> preset shows both. |
| 3D model file             | Your own <code>.gltf</code> , <code>.glb</code> or <code>.obj</code> . It is centred and scaled to fit automatically, and uses the layer's material.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Copies (instancing)

One layer can show many copies of its shape. Pick a **Layout**. On desktop and in WebGPU browsers the graphics card places the copies, variation included, so even big counts cost the processor almost nothing; in WebGL2 browsers the processor does the same maths and gets the same picture, just slower. (Copies on a shape's surface or on a terrain are placed by the processor everywhere, then finished on the graphics card.)

| Layout      | Settings                                                                                     | Example                                                                                                                                                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Single      | n/a                                                                                          | One hero object.                                                                                                                                                                                                                                                         |
| Grid        | Count x/y/z, Spacing                                                                         | Speaker stacks, cube cities.                                                                                                                                                                                                                                             |
| Radial      | Count, Radius. Copies face outward.                                                          | Panels around an arena, crystal skyline.                                                                                                                                                                                                                                 |
| Scatter     | Count, Radius, Only surface, Seed                                                            | Floating debris, asteroid shells.                                                                                                                                                                                                                                        |
| Orbit swarm | Count, Radius, Spread, Orbits / loop, Seed                                                   | Debris orbiting a planet.                                                                                                                                                                                                                                                |
| Big swarm   | Form (Orbits, Cloud, Shell, Galaxy), Count up to 250,000, Radius, Spread, Turns / loop, Seed | Huge swarms, placed by the graphics card on desktop and in WebGPU browsers (in WebGL2 browsers the processor places them, the same picture but slower, so keep counts lower there). Use a simple shape such as a tetrahedron. The <b>Galaxy Swarm</b> preset has 40,000. |
| Wall        | Columns, Rows, Spacing, Curve                                                                | Orb walls, LED walls. Curve bends the wall into an arc.                                                                                                                                                                                                                  |

|                      |                                                                                      |                                                                                                                                                                                           |
|----------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Spiral               | Count, Radius, Height, Turns                                                         | Chandeliers, DNA helices.                                                                                                                                                                 |
| Along a curve        | Curve and Frequencies (the ribbon curves), Size, Count, Laps / loop, Face along      | Beads riding a neon ribbon (give both the same curve, size and position), trains, conga lines.                                                                                            |
| On a shape's surface | On shape, Shape size, Count, Seed, Stand up, Lift. Copies are spread evenly by area. | Spikes on a crystal, moss on rocks. Set <b>Shape size</b> to the other layer's Size and put both layers in the same place.                                                                |
| On a terrain         | Terrain (a Terrain layer), Count, Seed, Follow the slope, Lift                       | Trees, crystals, lamps and ruins standing on the landscape. They ride along as it scrolls, follow its placement and shrink away at its edges where it wraps; on water or lava they float. |

**Variation** (appears when there are copies) adds life:

- **Random tilt**, **Random size**, **Random hue** and **Random spin** make every copy a little different. Change **Seed** to reshuffle them.
- **Size wave** is a wave of growing and shrinking that travels across the copies.
- **Light chase** is a bright glow running along the copies, like marquee lights.
- **Waves / loop** sets how many times those waves pass per loop, and **Wavelengths** sets how many waves fit across all copies.

**Colours across copies** (also when there are copies) spreads 2–4 colours over them: **Gradient** blends smoothly (back to the first colour at the end), **Steps** gives each copy one colour in turn. **Travel / loop** makes the colours run along the copies; **Colour the glow** tints the glow too.

## Deform

Bends the shape itself, on every copy. All values have ~, so they can move with the beat.

| Setting | Meaning                                                                                                                                                        |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Twist   | Turns of twist from the bottom of the shape to its top.                                                                                                        |
| Bend    | Curves the shape into an arc (degrees from bottom to top).                                                                                                     |
| Taper   | Top wider (+) or narrower (–) than the bottom.                                                                                                                 |
| Wobble  | Bumps flowing over the surface; <b>Bump size</b> and <b>Flow / loop</b> tune them. Simple shapes need <b>Subdivide</b> (in Relief) for smooth bends and bumps. |
| Explode | Faces fly apart, clearest with flat shading. Try ~ → Exp fade out, once per beat.                                                                              |

## Placement & motion

| Setting            | Meaning                                              |
|--------------------|------------------------------------------------------|
| Position, Rotation | Where the layer sits and how it is turned (degrees). |

|             |                                                                                                   |
|-------------|---------------------------------------------------------------------------------------------------|
| Size        | The size of the object, or of <b>each copy</b> . It does not change the distances between copies. |
| Stretch     | Stretch per axis (make a cube into a slab or a pillar).                                           |
| Spin / loop | Whole turns per loop around x, y and z. Negative numbers spin the other way.                      |
| Bob         | Up/down offset. Animate it with ~ to make objects float.                                          |

## 5.4 Symmetry

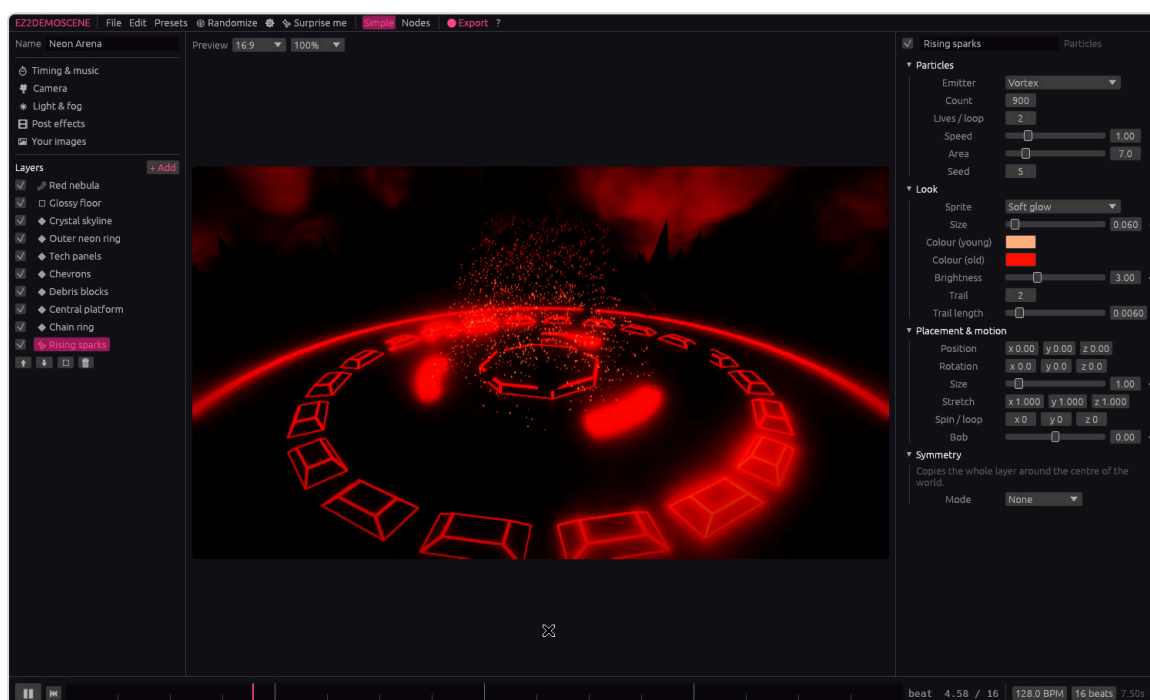
Symmetry copies the **whole layer** around the centre of the world:

- **Mirror X** (left/right), **Mirror Z** (front/back), **Mirror XZ** (four copies).
- **Radial**: N rotated copies around the vertical axis.
- **Kaleido**: N copies where every other one is mirrored, giving a kaleidoscope pattern.

### Tip

Build one side of a room, then add **Mirror X**. The Gold Kaleido Room's orb walls, speakers and light panels are each built once and mirrored.

## 5.5 Particles



A particle layer.

| Emitter      | Motion                                                  |
|--------------|---------------------------------------------------------|
| Burst        | Explodes outward from the centre.                       |
| Sphere drift | Floats gently inside a sphere.                          |
| Ring orbit   | Circles in a flat ring (Speed = whole orbits per loop). |

|                |                                                                                                                                                                    |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fountain       | Shoots up and falls back.                                                                                                                                          |
| Warp stars     | Streams towards the viewer (hyperspace). Add a <b>Trail</b> for streaks.                                                                                           |
| Vortex         | Spirals upwards like embers.                                                                                                                                       |
| Snow / glitter | Falls slowly and sparkles.                                                                                                                                         |
| Tornado        | Spirals up a swaying funnel that widens with height, with debris circling its foot. Use it with <b>Smoke</b> for a dark twister, or bright blue for a water spout. |

**Smoke** makes particles cover what is behind them instead of adding light, so they can be dark (smoke, dust, a tornado). **Brightness** then becomes **Opacity**.

**Count** is the number of particles. **Lives / loop** is how many times each particle is reborn in one loop, so higher means faster. **Area** is the size of the emitter. The look is controlled by **Sprite** (soft glow, pixel square, star, ring), **Size**, two colours (young → old), **Brightness** and **Trail** (ghost copies behind each particle).

### Performance

Particles are cheap. Tens of thousands are fine on a normal GPU. Trails multiply the count, so  $count \times (trail + 1)$  is what gets drawn.

## 5.6 Terrain

An endless landscape that rushes towards the camera, in the retro vector style. Add it with **+ Add** →  **Terrain**.

| Setting                                         | Meaning                                                                                                                                                                                                                                                        |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Style                                           | <b>Wireframe</b> (glowing grid lines only; things behind show through), <b>Solid</b> (lit ground) or <b>Solid + lines</b> .                                                                                                                                    |
| Height, Hills, Roughness                        | How tall the mountains are, how many hills fit across the terrain, and how bumpy they are.                                                                                                                                                                     |
| Valley                                          | Flattens a road down the middle (0 = none), the classic synthwave look.                                                                                                                                                                                        |
| Scroll / loop                                   | How many times the landscape scrolls past per loop. It repeats exactly, so the loop stays seamless. 0 stands still.                                                                                                                                            |
| Line colour, Line glow, Ground colour           | The look. Animate Line glow with a Pulse for beat flashes.                                                                                                                                                                                                     |
| Texture, Tiles, Texture on lines, Chunky pixels | Maps a texture onto the ground (and optionally the grid lines). It moves with the landscape; <b>Tiles</b> is how many times it repeats across the terrain, a whole number so the loop stays seamless.                                                          |
| Size, Grid cells                                | Its extent and resolution. More cells is smoother but slower. Grid lines sit on every cell.                                                                                                                                                                    |
| Level of detail                                 | Full resolution near the camera, gradually coarser further away, with no visible steps: a quarter of the triangles. Raise <b>Grid cells</b> (up to 512) or <b>Size</b> for sharper or bigger landscapes at the same cost. Distant ridges get slightly simpler. |

|       |                                                                                                                                                                                                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shape | <b>Hills</b> (rolling), <b>Ridged mountains</b> (sharp crests), <b>Mesas</b> (flat terraces with cliffs), <b>Sand dunes</b> (gentle on one side, steep on the other), <b>Canyons</b> (a plateau cut by winding gorges) and <b>Craters</b> .                                                                                           |
| Biome | Colours solid ground by height and steepness instead of one <b>Ground colour</b> : <b>Alpine</b> (grass, rock, snowy peaks, sandy shores), <b>Desert</b> (sand and banded red cliffs), <b>Volcanic</b> (black basalt with glowing cracks), <b>Arctic</b> (snow and blue ice), <b>Alien</b> (purple moss, teal crystal, glowing tips). |

## Water & lava

A **Liquid** fills everything below its **Level** with a flat surface. The level is a fraction of the mountain height; animate it for tides or rising lava. Liquids also show up on wireframe terrain.

| Liquid    | Look                                                                                                  | Shine / Glow, Ripples / Crust... |
|-----------|-------------------------------------------------------------------------------------------------------|----------------------------------|
| Water     | Reflects the sky with a sun glint, is lighter in the shallows and has foam and wet sand at the shore. | Shine, Ripples                   |
| Lava      | Glowing rivers under a dark, cracked crust that drifts and churns. It lights up the nearby rocks.     | Glow, Crust                      |
| Toxic goo | Glowing green slime with popping bubbles.                                                             | Glow, Bubbles                    |
| Ice       | A frozen, cracked, shiny surface that stays still.                                                    | Shine, Cracks                    |

**Current / loop** makes the surface drift along the terrain a whole number of times per loop; the ripples and crust also turn gently once per bar. Both loop exactly.

## 5.7 Laser beams

Club-style beams shooting from the layer's position along its up axis. Turn the layer with **Rotation** to aim them, and use **Mirror X** symmetry for a pair of laser rigs.

| Setting                      | Meaning                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pattern                      | <b>Fan</b> (a flat fan that sweeps side to side), <b>Rotating cone</b> or <b>Scatter</b> (random directions that wobble).                                                                                                                                                                                                                                     |
| Beams, Spread, Length, Width | How many beams, how wide the fan or cone opens (degrees), and their size.                                                                                                                                                                                                                                                                                     |
| Colour (first/last)          | The beams blend from the first colour to the last.                                                                                                                                                                                                                                                                                                            |
| Sweep, Sweeps / loop         | How far the beams swing, and how often per loop (for the cone, also how fast it turns).                                                                                                                                                                                                                                                                       |
| Beat strobe                  | Makes the beams flash on every beat and fade until the next one.                                                                                                                                                                                                                                                                                              |
| Style                        | <b>Laser beams</b> or <b>Spotlight cones</b> : wide, hazy cones of light, brightest where you look through the most haze. Set their <b>Cone angle</b> , and tick <b>Light pools</b> for bright spots where the cones hit the ground (height 0). Point the layer down (Rotation x 180) to hang the lamps from a ceiling. Add some <b>Mist</b> for a club look. |

## 5.8 Neon ribbon

A glowing tube along a closed curve, with pulses of light running along it. Choose a **Curve** (**Lissajous**, **Knot**, **Figure eight**, **Wavy ring**, **Rose**) and its wave counts, then set **Pulses**, **Laps / loop** (negative runs backwards), **Pulse length** and **Pulse glow**. Ribbons reflect in mirror floors and can take symmetry.

## 5.9 Weather

+ **Add** → 🌧️ **Weather** fills a box around the camera with rain, snow, embers, sand or fireflies. The box follows the camera, so the weather is everywhere you look. Its **Ground height** is where rain splashes and embers start.

| Setting                          | Meaning                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kind                             | <b>Rain</b> (streaks and splash rings), <b>Snow</b> (swaying flakes), <b>Rising embers</b> (flickering sparks), <b>Sandstorm</b> (dust blown along the wind), <b>Fireflies</b> (wandering, blinking lights) or <b>None</b> (lightning only).                                                                                                    |
| Amount, Size, Colour, Brightness | How many drops and how they look. Rain also has <b>Streak length</b> and <b>Splashes</b> .                                                                                                                                                                                                                                                      |
| Falls / loop                     | How many times each drop falls per loop: higher is faster. It is a whole number, so the weather loops.                                                                                                                                                                                                                                          |
| Wind, Wind direction             | Slants the rain and snow. Animate Wind with a Sine for gusts.                                                                                                                                                                                                                                                                                   |
| Area, Height                     | The size of the box around the camera.                                                                                                                                                                                                                                                                                                          |
| Wet ground / Snow cover          | Rain makes every surface darker and glossier, with puddles on flat ground that mirror the sky and ripple with rain rings. Snow settles on everything facing up. A snow cover can't really build up over a loop (it would have to jump back at the end), but you can animate it, for example with a <b>Swell</b> so it builds up and melts away. |
| Lightning                        | Jagged bolts that strike in front of the camera and a flash that lights up the whole scene, clouds included. <b>Chances / loop</b> and <b>Chance</b> decide how often; each strike flickers two or three times.                                                                                                                                 |

### Performance

Weather drops cost about as much as particles. Volumetric clouds are the heavy part of a storm scene: lower their **Quality** on phones.

## 5.10 Waterfall

+ **Add** → 🌊 **Waterfall** adds a curtain of **water**, **lava** or **toxic goo** that pours from the layer's position downwards and arcs out along its Z axis, with streaks running down it and foam, spray or smoke at its foot. Place it on the edge of a cliff: a terrain mesa, a shape or a wall of boulders (see the Rainbow Falls preset).

| Setting            | Meaning                                                        |
|--------------------|----------------------------------------------------------------|
| Width, Height, Arc | The size of the curtain and how far it arcs out from the edge. |

|                     |                                                                             |
|---------------------|-----------------------------------------------------------------------------|
| Flow / loop         | How many times the streaks run down per loop (a whole number, so it loops). |
| Brightness / Glow   | Water brightness, or the glow of lava and goo.                              |
| Foam & mist / Smoke | Puffs at the foot (0 = none).                                               |

## 5.11 Text

Glowing letters in the scene, on a flat sign facing forward (tick **Face the camera** to keep them readable from any side). They stay sharp at any size. Add one from **+ Add →**  **Text**.

| Style          | What it does                                                                                                                                          |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Still          | The text, centred. Several lines are fine.                                                                                                            |
| Scroller       | Runs across a <b>Window</b> a whole number of times per loop ( <b>Runs / loop</b> ; negative runs the other way). Letters fade at the window's edges. |
| Sine scroller  | A scroller riding a sine wave: <b>Wave height</b> , <b>Wave length</b> (letters per wave) and <b>Wave rolls / loop</b> .                              |
| Typewriter     | Letters appear one by one, <b>Letters / beat</b> , starting again with the loop.                                                                      |
| Greetings list | One line at a time, fading in and out, a new line every <b>Beats / line</b> (4 = one per bar).                                                        |

**Font & look:** Pixel (chunky demoscene letters), Mono or Sans, or your own TTF/OTF file; size and spacing; a top and a bottom colour; **Glow** (brightness, above 1 it blooms; it has ~); an outline; a drop shadow; and **Chrome**, which bevels the letters and reflects the sky like a 90s logo. The **Oldschool Intro** preset uses all three styles.

## 5.12 Sprites

Pictures in the scene, one per copy: glows, flames, sparkles, coins, a logo on a sign. Add them from **+ Add →**  **Sprites** (Glow dots, Flames, Explosion or Image plane).

| Setting                   | What it does                                                                                                                                                                                                                        |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Image                     | A built-in texture, one of your images, or none (a soft glowing dot). The built-in <b>sheet_...</b> images are animated sprite sheets: explosion (plays once), flame, spinning coin and sparkle (these loop).                       |
| Facing                    | <b>Face the camera</b> (always square on), <b>Upright</b> (stands up and only turns around, like trees or flames) or <b>Fixed plane</b> (a flat picture you place and turn like any layer).                                         |
| Blend                     | <b>Alpha:</b> soft edges from the image's transparency (copies are drawn far to near). <b>Additive:</b> light adds up, for glows and fire. <b>Cutout:</b> hard edges at half transparency, solid like a shape (best for pixel art). |
| Size, Opacity, Tint, Glow | Height (the width follows the picture), transparency, a colour multiplied in, and brightness (above 1 it blooms). All but Tint have ~ and  .   |

|           |                                                                                                                                                                                                                                                                                                                                                              |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Animation | For a sprite sheet (frames in a grid, left to right, top to bottom): <b>Columns</b> , <b>Rows</b> , <b>Frames</b> used (0 = all) and <b>Plays / loop</b> , a whole number so the animation lines up with the loop. <b>Each copy starts on its own frame</b> keeps a crowd of flames from flickering in step. Picking a built-in sheet sets the grid for you. |
| Copies    | The same copy layouts and variation as shapes (grid, ring, scatter, orbit, on a curve, on a terrain...).                                                                                                                                                                                                                                                     |

The **Campfire Sprites** preset uses flames, sparkles and spinning pixel coins.

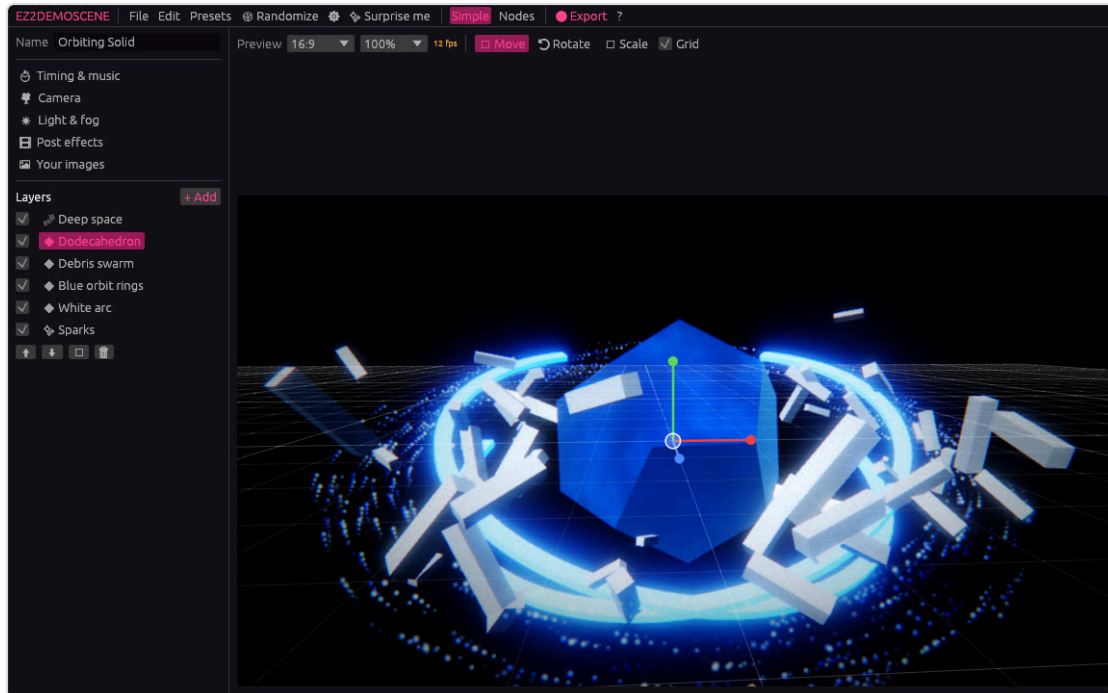
## 5.13 Electric arcs

Tesla-coil lightning, from **+ Add → ⚡ Electric arcs**. Each arc zigzags between its two ends, crawls while it lasts and takes a new shape at every strike.

| Setting                             | What it does                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path                                | <b>Between two points</b> ( <b>From</b> and <b>To</b> , placed with the layer), <b>To the nearest copies</b> of a shape or sprite layer ( <b>Target</b> ; from this layer's position to its <b>Arcs</b> nearest copies, so they jump as the copies move) or <b>Copy to copy</b> (from each copy of the target to the next, round the ring). |
| Strikes / loop                      | How many new shapes per loop, a whole number so the loop closes.                                                                                                                                                                                                                                                                            |
| Jaggedness, Crawl, Fade             | How far the arc zigzags; how far the zigzag slides during a strike; how much each strike dims before the next (0 = steady).                                                                                                                                                                                                                 |
| Branches, Width, Colour, Glow, Seed | Side forks, thickness, colour, brightness (~ and 🎵: flash it on the kick) and a different set of zigzags.                                                                                                                                                                                                                                   |

The **Tesla Swarm** preset strikes the nearest debris from the core and runs arcs round a ring of coils.

## 5.14 Placing objects in the viewport



The Dodecahedron is selected: the Move handles appear on it, and the ground grid is on.

- **Select** a shape or particle layer by clicking it in the viewport, or in the layer list.
- Pick a tool above the viewport: **Move** (**W**), **Rotate** (**E**) or **Scale** (**R**).
- **Drag a handle.** Red is X, green is Y (up) and blue is Z. With Move, the white circle in the middle slides the object along the floor. With Rotate, drag one of the three rings. With Scale, the coloured squares stretch one axis and the middle circle changes the overall size.
- Hold **Ctrl** while dragging to **snap**: moves go in 0.25 steps, rotations in 15° steps and sizes in 0.1 steps.
- **Grid** (**G**) shows the ground grid, where one square is one unit.
- Dragging anywhere else still turns the camera, and the mouse wheel still zooms.

### Tip

Pause first (**Space**) when placing spinning or orbiting things. Every handle drag counts as a single step for **Ctrl** + **Z**.

## 6 Materials, glow & textures

### Surface

| Setting   | Effect                                                                                                                      |
|-----------|-----------------------------------------------------------------------------------------------------------------------------|
| Colour    | The base colour.                                                                                                            |
| Metallic  | 0 looks like plastic or paint. 1 looks like metal, where reflections take on the colour (gold, chrome, copper).             |
| Roughness | 0 is a glossy mirror-like finish, 1 is matte. The glossy black look of Neon Arena is black + metallic 0.9 + roughness 0.15. |
| Faceted   | Flat-shaded, low-poly look. Every face catches light separately.                                                            |
| Rim light | A glow along the silhouette that separates objects from dark backgrounds.                                                   |

### Glow (emission)

**Glow colour** and **Glow** (strength) make a surface emit light. Bright glow feeds the **bloom** post effect and produces the neon halo. **Glow where** decides which part glows:

| Glow where | Result                                                                                                                              |
|------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Full       | The whole surface glows (neon tubes, light panels).                                                                                 |
| Edges      | Only the outlines of each face glow, for a Tron / wireframe-neon look. Works on the solids, panels, cylinders, shards and pyramids. |
| Stripes    | Glowing horizontal stripes.                                                                                                         |
| Texture    | Bright parts of the texture glow (LED grids, circuits, tech-panel slits).                                                           |

**Hue shift** rotates all the colours of the layer. Animate it with a Saw wave for rainbow cycling.

### Relief: bump, normal and displacement maps

The **Relief** section of a shape layer makes its surface uneven. It uses the material's **Tiling** and **Scroll**, and the colour texture unless you pick a separate **Relief texture**.

| Setting          | Result                                                                                                                                                                 |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mode: Bump       | The texture's brightness is height: bright parts look raised, dark parts sunken (mortar between bricks, rivets, grid lines). Only the lighting changes, so it's cheap. |
| Mode: Normal map | The texture is a normal map (the purple-blue images from 3D tools and texture sites). Import one as an image and pick it as the relief texture.                        |
| Relief           | Strength of the bump or normal map (animatable).                                                                                                                       |
| Displacement     | Really moves the surface outwards by the texture brightness (negative pushes inwards). Animate it for pulsing, breathing shapes.                                       |

Subdivide

Adds geometry detail (each level splits every triangle in four), so displacement can show fine patterns.

### Tip

Displacement looks best on smooth shapes (sphere, torus, capsule, rounded cube). Faceted shapes such as the cube or the gem open up at their edges, because each face moves along its own direction.

## Glitch

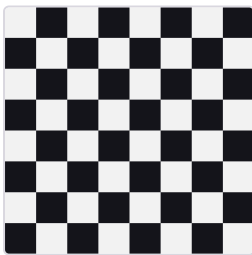
The **Glitch** section of a shape layer corrupts its geometry, like a broken signal. Set an **Amount** and a **Style**:

| Style        | Result                                                                     |
|--------------|----------------------------------------------------------------------------|
| Jitter       | The corners shake to random positions; the shape crumples without tearing. |
| Slices (VHS) | Horizontal bands of the object jump sideways.                              |
| Shatter      | The faces fly apart along their directions.                                |

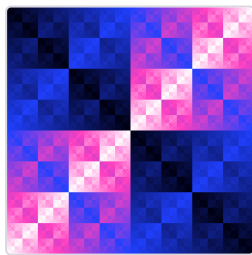
**Changes / loop** is how many new random patterns appear per loop, and **Chance** is how many of those moments actually glitch (1 = all the time, 0.3 = short bursts). Like everything else, the glitch pattern repeats exactly every loop. Animate **Amount** with a Pulse for hits on the beat.

## Textures

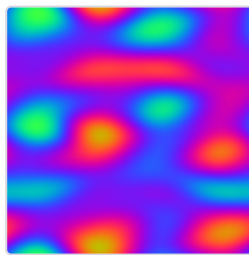
Pick a **Texture** from the built-in retro pack or from your own images. **Tiling** repeats it, **Scroll / loop** slides it by whole tiles per loop, and **Chunky pixels** keeps the pixels sharp instead of smoothing them.



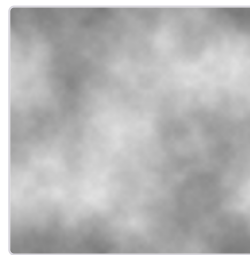
checker



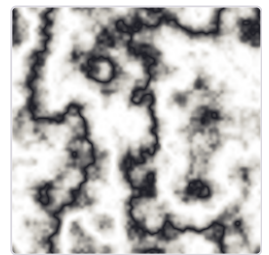
xor



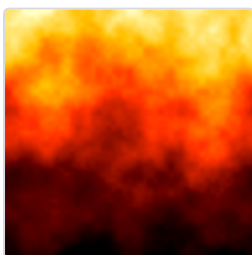
plasma



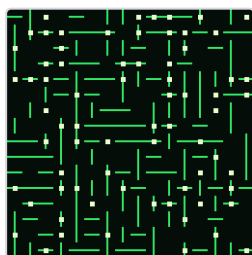
noise



marble



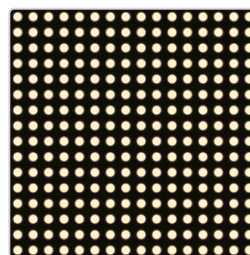
fire



circuit



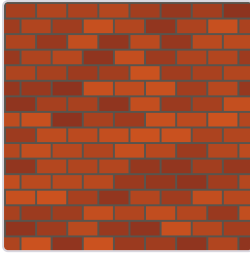
tech\_panel



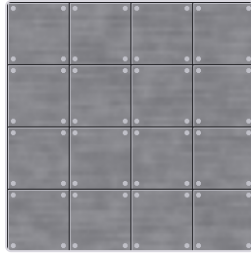
led\_grid



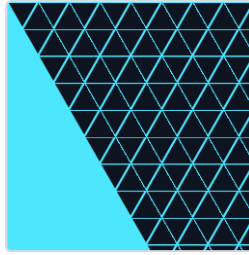
speaker



brick



metal\_plate



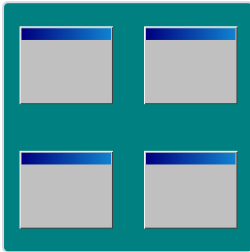
hex



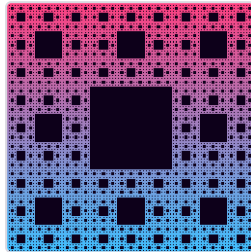
stripes



copper



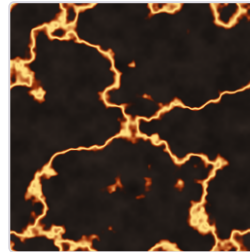
win9x



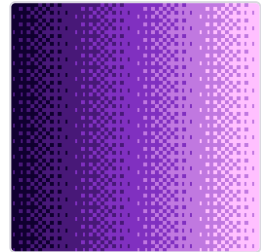
sierpinski



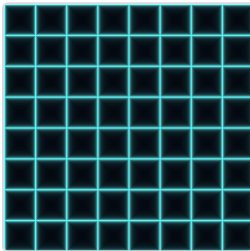
stars



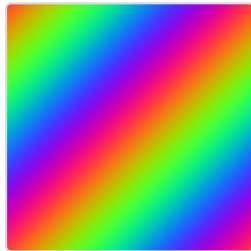
lava



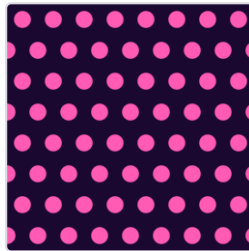
dither



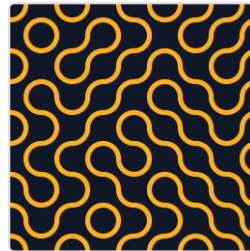
tron\_grid



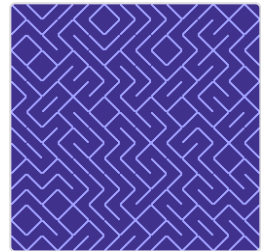
rainbow



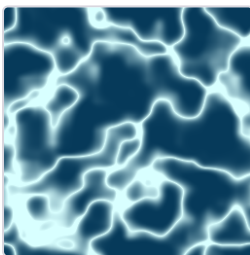
polka



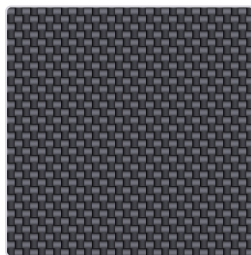
truchet



maze



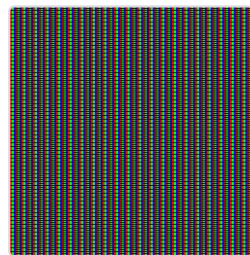
caustics



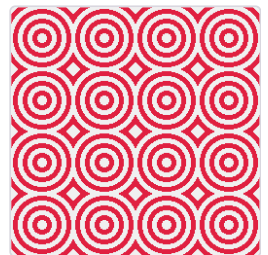
carbon



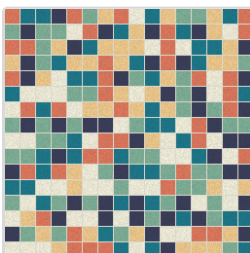
wood



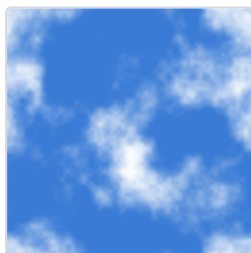
crt\_mask



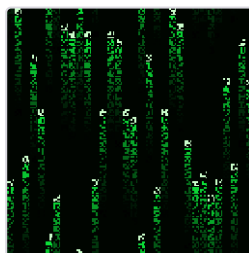
rings



mosaic



clouds



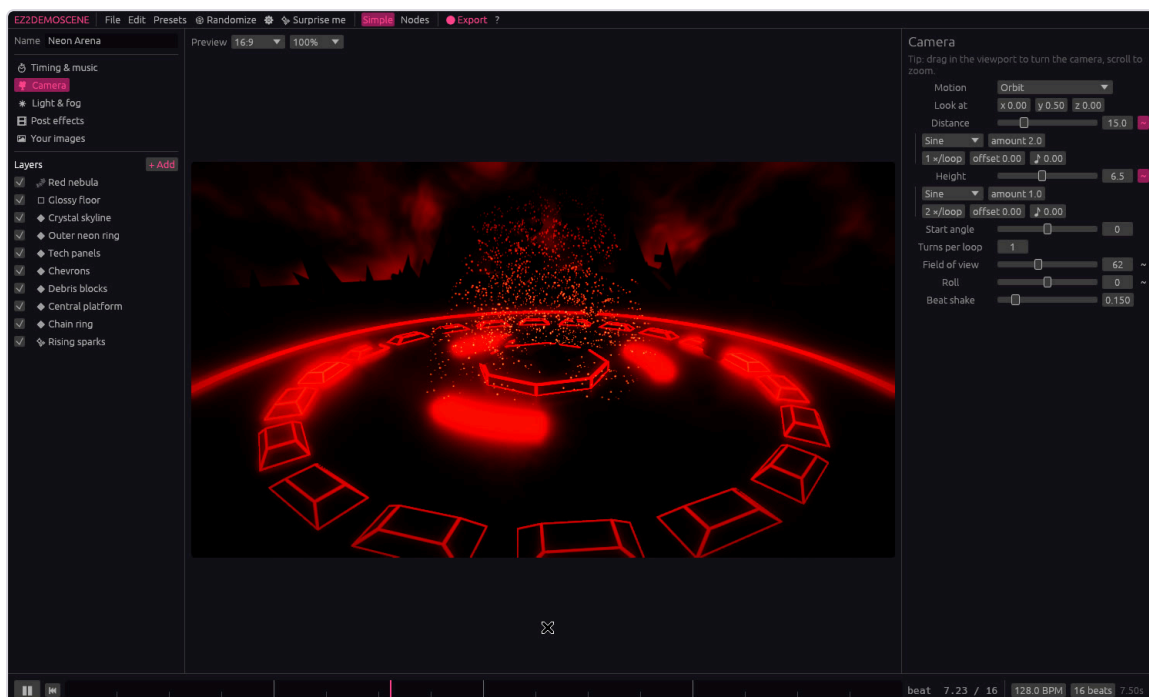
matrix

## Your own images & "retro-ize"

Import images (PNG, JPG, BMP, GIF, TGA) in any of these ways: from a texture menu (**Import image...**), from **File → Import image...**, or by dragging them onto the window. A dropped image is applied to the selected shape. All your images are listed under **Your images**, where you can tick **Retro-ize**. This shrinks the image to **Max size** pixels and remaps it to a classic **Palette** with **Dithering**, turning any photo into a 90s texture.

## 7 Animating anything

Every value with a  button next to it can move over time. The button turns pink when the value is animated. Click it to open the animation panel:



In the Camera section, Distance and Height are animated: their panels show wave, amount,  $\times/\text{loop}$ , offset and  $\text{J}$ .

| Control              | Meaning                                                                                                                                                                          |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wave                 | The shape of the movement, in three groups (see below). Hover a shape in the menu for a description.                                                                             |
| amount               | How far the value moves from the slider value. 0 means not animated. Picking a shape or a rhythm on a still value gives it a starting amount so you see something straight away. |
| $\text{J}$           | Rhythm menu: <b>twice per beat, every beat, every 2 beats, every bar, every 2 or 4 bars or once per loop</b> . It sets $\times/\text{loop}$ for you from the loop length.        |
| $\times/\text{loop}$ | Whole cycles per loop. The number of beats in the loop hits <b>every beat</b> . Negative numbers run backwards.                                                                  |
| offset               | Shifts the timing (0–1 of a cycle), to stagger several animations.                                                                                                               |
| $\text{J}$           | Adds the music's loudness (see chapter 10).                                                                                                                                      |

Below the controls, a small **graph** shows the value over the whole loop, with beat ticks and a dot at the current moment.

| Group | Shapes | Use them for |
|-------|--------|--------------|
|-------|--------|--------------|

|            |                                                                                         |                                                                                                                                                                                                          |
|------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LFO        | Sine, Triangle, Saw up, Saw down, Square                                                | Smooth or hard back-and-forth movement around the slider value.                                                                                                                                          |
| Beat fades | Pulse (hit), Exp fade in, Exp fade out, Linear fade in, Linear fade out, Swell in & out | Envelopes that run from 0 to the full amount (or back) once per cycle. Set J → <b>every beat</b> for hits that decay (Exp fade out), build-ups that drop on the beat (Exp fade in), or breathing swells. |
| Random     | Sample & hold, Smooth random, Drunken walk                                              | A new random value each cycle (held, or gliding smoothly), or a random walk that wanders and always finds its way back by the end of the loop.                                                           |

Almost every number can be animated this way, including **post effects** (bloom threshold, pixel size, dither, CRT scanlines and curvature, contrast, saturation, vignette, grain, beat flash, kaleidoscope angle), materials (metallic, roughness, rim light, tiling), light, camera swing and shake, particle speed and area, mirror reflectivity and blur, laser spread, length, width and sweep, terrain roughness and valley, and ribbon pulses.

## Recipes

- **Pulse on every beat:** Glow → Pulse, amount 3,  $\times/\text{loop} = \text{loop beats}$  (e.g. 16).
- **Breathing camera:** Camera Distance → Sine, amount 2,  $\times/\text{loop} 1$ .
- **Kick zoom:** Kaleidoscope Zoom → Pulse, amount 0.2,  $\times/\text{loop} 16$ .
- **Rainbow:** Hue shift → Saw, amount 0.5,  $\times/\text{loop} 1$ .
- **Floating object:** Bob → Sine, amount 0.5,  $\times/\text{loop} 2$ .
- **Strobe:** Glow → Square,  $\times/\text{loop} 32$ .
- **Beat hit:** Glow → Exp fade out, J every beat.
- **Build-up:** Bloom threshold → Exp fade in, J every 4 bars (negative amount lowers it towards the drop).
- **Nervous camera:** Camera Height → Drunken walk, J every beat, amount 1.

## Shake

**Placement & motion** → **Shake** jolts a layer in a random direction. **Shake distance** and **Shake turn** set how far, and **New direction** how many times per loop a new random direction is picked. Give the distance a fade for a hit that settles: the  **Beat jolt** button does exactly that (Exp fade out, every beat). The **Jitter** node can drive the same shake for everything connected to it (chapter 12).

## Blink / strobe

Every layer has a **Blink / strobe** section that makes the **whole layer** appear and disappear, or flash:

| Mode         | What it does                                                                                   |
|--------------|------------------------------------------------------------------------------------------------|
| Blink        | On and off in a regular rhythm. <b>On time</b> is the fraction of each blink it stays visible. |
| Random blink | On and off at random moments. <b>Chance on</b> sets how often it is visible.                   |

---

Flash

Stays visible but its glow flashes and fades. **Dim level** is the brightness between flashes.

---

**Times / loop** sets the rhythm (16 = every beat of a 16-beat loop) and **Offset** shifts it, so two layers can blink in turn.

## 8 Camera, light & fog

### Camera

| Setting          | Meaning                                                                                                                                                                                                                                                                  |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Motion           | <b>Orbit</b> circles the target <b>Turns per loop</b> times. <b>Pendulum</b> swings back and forth by <b>Swing</b> degrees. <b>Static</b> stays put, though its values can still be animated. <b>Path</b> flies through your own points (see <i>Camera paths</i> below). |
| Look at          | The point the camera aims at.                                                                                                                                                                                                                                            |
| Distance, Height | How far away and how high the camera is. Both are animatable.                                                                                                                                                                                                            |
| Start angle      | Which side the camera starts from.                                                                                                                                                                                                                                       |
| Field of view    | Zoom lens: 20–35 for a telephoto look, 60–90 for wide and dramatic.                                                                                                                                                                                                      |
| Roll             | Tilts the horizon. Animate it for a dizzy rave feel.                                                                                                                                                                                                                     |
| Beat shake       | A small camera kick on every beat.                                                                                                                                                                                                                                       |
| Punch-in         | With a song loaded, the camera zooms in on every kick (or snare, hi-hat, any hit, note) and eases back.                                                                                                                                                                  |

#### Mouse

Dragging in the viewport changes **Start angle** and **Height**. Scrolling changes **Distance**. A Path camera flies by itself, so dragging does nothing then.

### Camera paths

Pick **Motion** → **Path** and the camera flies a smooth closed loop through your points, at an even speed, **Laps / loop** times per loop. Each point has a position (**Eye**), a **Look at** point, a field of view and a roll, and the camera blends between them. With the Camera selected, the viewport shows the flight as a yellow line with numbered points.

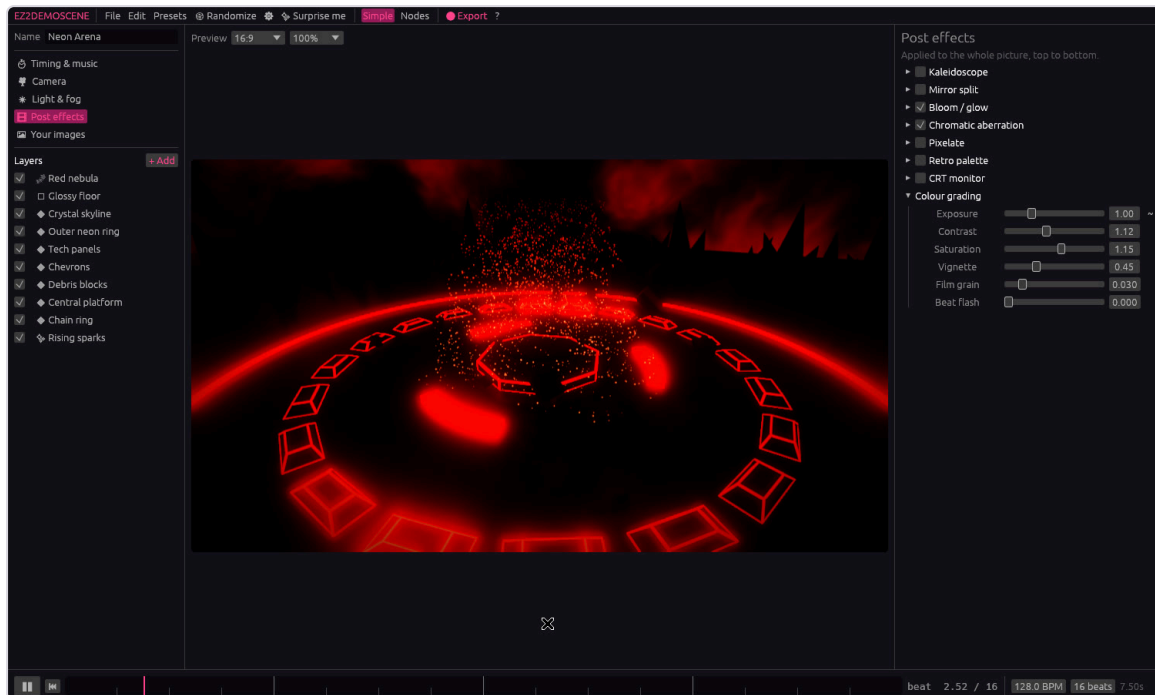
- **Adding shots:** switch to **Static**, frame the shot by dragging and scrolling in the viewport, then press  **Add this view to the path**. Repeat, then switch back to **Path**.
- **Changing a shot:** press  on a point to look from it (the camera switches to Static there), adjust it in the viewport, then press  on that point to store the new view.   reorder points,  removes one.
- **Linger** slows the camera down at every point: 0 is an even glide, 1 holds each shot for a moment.
- **Cut on hits:** with a song loaded, the camera cuts to the next point on every kick (or snare, hi-hat...) and **Drifts** a little towards the following one. The cuts restart with the loop, so the video still loops. Without music the camera just flies.

### Light & fog

- **Fog colour / Fog density:** distant things fade into this colour. The fog colour is also the "empty space" colour, so match it to your background.

- **Sky light / Ground light / Ambient:** soft light from above and below. The sky colour is also what glossy surfaces reflect, which gives black gloss its highlights.
- **Sun colour, Sun strength, Sun direction:** the main light.
- **Sun shadows:** shapes and terrain cast shadows from the sun (the moon at night) onto shapes, terrain, water and the mirror floor. **Darkness, Softness** and **Distance** (how far around the camera's target shadows reach: smaller is sharper). In dark neon scenes shadows are subtle, because most light comes from glow, not the sun.
- **Contact shadows:** soft dark patches under shapes standing on a mirror floor, so they don't look like they float.
- **Mist (height fog):** fog that is thick near the ground and thins out higher up, so it pools in valleys and over water. **Base height** is where it is densest, **Thickness** how high it reaches. It also hides the horizon of the background.
- **Underwater caustics:** the rippling web of light you see at the bottom of a pool, on every surface. **Only below** limits it to things under a water line.
- **Rainbow:** a rainbow (with a faint second bow) opposite the sun. Like a real one, it needs the sun fairly low: with the sun high above, the bow sinks below the horizon.
- **Day & night cycle:** the sun rises, crosses the sky and sets a whole number of times per loop. The light, fog and sky turn warm at sunrise and sunset; at night the sky darkens, stars come out, the moon rises opposite the sun and moonlight takes over. **Sun direction** sets where the sun is at noon. Clouds, god rays and reflections all follow the sun.

## 9 Post effects



Post effects. Tick a section's box to enable it, and click its arrow to open its settings.

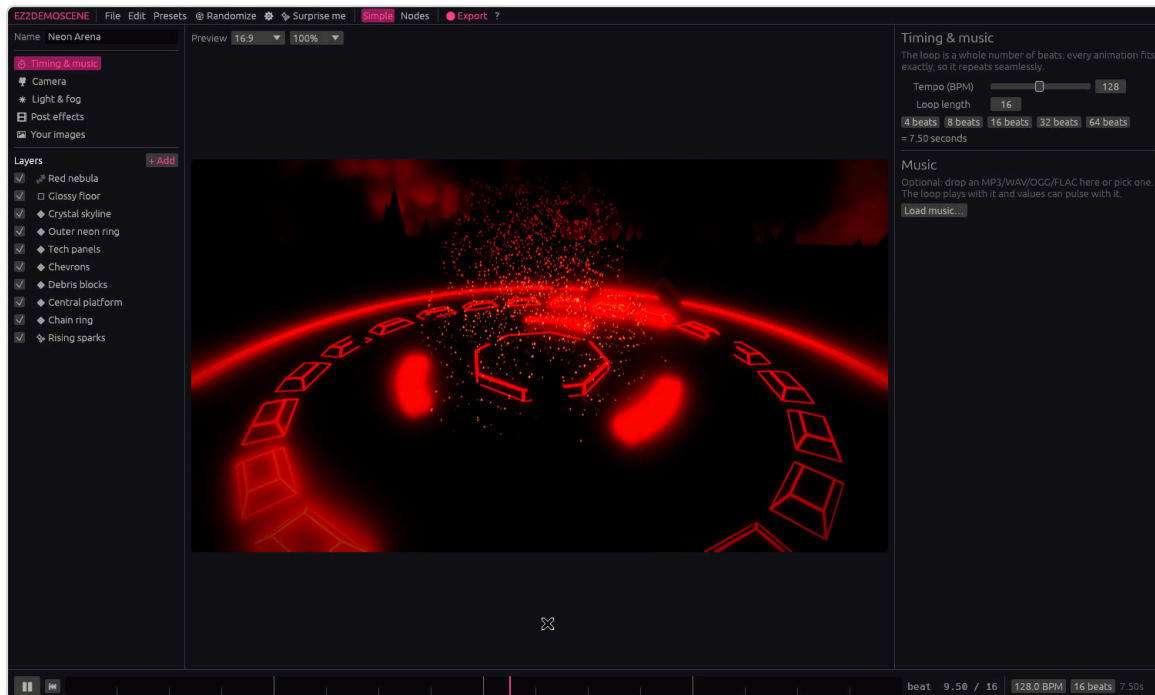
| Effect                | What it does                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kaleidoscope          | Folds the whole picture into mirrored wedges. You can set Segments, Angle, Turns per loop, Zoom (animatable) and Centre.                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Mirror split          | Mirrors one half of the picture onto the other (left → right, top → bottom, or quad). This is instant symmetry.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Bloom / glow          | Bright things spill light. Intensity sets the strength, Threshold how bright something must be to glow, and Spread how wide the halo is.                                                                                                                                                                                                                                                                                                                                                                                                                            |
| God rays & lens flare | Shafts of light streaming from the sun (or from the picture centre, for "light at the end of the tunnel"). Bright sky near the light is smeared towards it, so clouds, mountains and shapes in front cast streaks of shadow. Length, Threshold and Tint shape the rays; <b>Lens flare</b> adds ghosts and a streak when you look into the light, and fades when the sun is hidden.                                                                                                                                                                                  |
| Depth of field        | A camera lens blur: things nearer or further than the focus go soft, with round bokeh. <b>Auto focus</b> keeps the point the camera looks at sharp (or set a <b>Focus</b> distance); <b>Blur</b> sets how soft it gets. Both have ~ and 🎵, so the focus can pull or the blur can breathe on the kick.                                                                                                                                                                                                                                                               |
| Feedback trails       | Each frame keeps a fading copy of the previous ones, so moving things leave echoes. <b>Trail length</b> (with ~ and 🎵) sets how long they linger; <b>Zoom / second</b> makes the echoes fly out (above 1) or fall in, <b>Turn / second</b> spins them and <b>Colour drift</b> changes their hue as they fade. The speeds are per second, so they look the same at any frame rate. The preview builds trails as it plays and starts fresh after a jump; exports render one loop first (not saved), so frame 0 already has its trails and the loop closes seamlessly. |

|                      |                                                                                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heat haze            | The picture shimmers as if seen through rising hot air: above hot, bright things (lava, fire, the sun), near the bottom of the picture (hot ground) or everywhere. <b>Rises / loop</b> sets its speed. |
| Chromatic aberration | An RGB colour split towards the edges, like a cheap lens. Pulse it on the beat.                                                                                                                        |
| Pixelate             | Fat pixels (the size is measured at 1080p, so previews match exports).                                                                                                                                 |
| Retro palette        | Reduces the image to a classic computer palette with ordered (Bayer) dithering.                                                                                                                        |
| CRT monitor          | Scanlines, screen curvature and a VHS wobble.                                                                                                                                                          |
| Colour grading       | Exposure, Contrast, Saturation, Vignette (dark corners), Film grain and Beat flash (a white flash on every beat).                                                                                      |

## Palettes

| Palette            | Colours                                                                              |
|--------------------|--------------------------------------------------------------------------------------|
| EGA (16)           |    |
| CGA (4)            |     |
| Commodore 64 (16)  |   |
| Game Boy (4)       |   |
| PICO-8 (16)        |  |
| Amiga copper (16)  |  |
| ZX Spectrum (8)    |   |
| VGA cube (216)     | Every colour quantised to 6 levels per channel, a subtle retro banding.              |
| Green phosphor (4) |   |

# 10 Timing & music



Timing & music.

## Tempo and loop length

**Tempo (BPM)** and **Loop length** (in beats) set how long the loop is. For example,  $128 \text{ BPM} \times 16 \text{ beats} = 7.5 \text{ seconds}$ . The quick buttons set 4, 8, 16, 32 or 64 beats. These values also appear in the timeline.

Because every animation is counted per loop, **changing the BPM speeds the whole scene up or slows it down**, and it stays in sync.

## Adding music

Use **Load music...** or drop an MP3, WAV, OGG, FLAC, M4A or AAC file onto the window. The song is **analysed** once when it loads: loudness, kick, bass, mids, highs and brightness curves; the **hits** (kicks, snares and claps, hi-hats, any onset, new notes); a 16-band spectrum; the melody's note; and the **tempo**. The timeline shows the loudness, with kicks as red ticks and snares as yellow ones. The **Meters** show every source live. The analysis runs in the background (**Analysing music... 42%** in the viewport bar), so you can keep editing; music-driven settings start reacting when it finishes, and exports wait for it. Changing only the MIDI file or its offset reuses the analysis instead of starting again.

## Loop a part of the song, or the whole song

- **Loop a part of the song** (default): the loop reacts to a window of the song as long as the loop, starting at **Starts at**. The window is treated as a circle: the last moments fade into its start, and a hit near the end keeps ringing after the loop point, so the loop stays seamless. Video exports contain exactly that window of the song, repeated. **Use song tempo** sets the BPM to the

detected tempo and snaps the window onto a bar; **Snap to bar** only snaps; **Fit whole song** makes the whole track one loop.

- **Whole song**: the preview and exports run from the start to the end of the song (repeats don't apply); loop animations keep cycling underneath.

## Making values react: the 🎵 row

Every ~ panel has a 🎵 row under the rhythm settings. Pick a source and an amount:

| Source                                                | What happens                                                                                                                                                                                       |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Follow: Loudness, Kick, Bass, Mids, Highs, Brightness | The value rises with that part of the sound. Tick <b>smooth</b> for a calmer curve and use <b>ignore below</b> to react only to the loud parts.                                                    |
| Follow: Melody pitch                                  | The melody's note from C (0) to B (11/12). On <b>Hue shift</b> with amount 1, each note gets its own colour.                                                                                       |
| Each kick / snare / hi-hat / hit / new note           | Plays a <b>shape</b> (a beat fade, e.g. Exp fade out) of the chosen <b>length in beats</b> on every hit, scaled by how hard it hit. Flashes on the song's real kicks rather than on the metronome. |

A negative amount pulls the value down instead (e.g. the camera distance on each kick, for a punch-in). The curve preview shows the result with the music.

## Time warp

**Time warp** makes motion (spins, orbits, camera moves, scrolling, clouds, LFOs) run faster while the chosen source is strong, so the scene surges on every kick or on the loud parts. The warped time is rescaled so a full loop still ends exactly where it started, so the loop stays seamless. Beat fades, strobos, blinks and shakes stay on the real beat. A negative amount slows motion down on hits.

## Equalizer copies

Shapes with copies have an **Equalizer** setting (under Copies → variation): each copy grows upwards and glows with one frequency band, low notes first. 16 copies in a row or an arc make a classic spectrum display (see the **Music Reactor** preset).

## MIDI notes

**Load MIDI notes...** (or drop a .mid file) takes the hits and the melody from a MIDI file instead of guessing them from the audio: drums on channel 10 give exact kicks, snares and hats, other channels give new notes and the melody pitch, with their velocities. **MIDI offset** lines the notes up with the audio. A MIDI file works without an audio file too.

## Live input

**Live input (microphone / line-in)** makes the preview react to what the microphone hears, for VJ sets. Every source works live, and so does time warp. Live input is not repeatable, so exports always use the music file.

### Tip for musicians

Load your track, press **Use song tempo**, pick a strong section with **Starts at**, then link the glow of your shapes to **Each kick**. Export with the music included and the visuals land exactly on your beats.

## Scenes & timeline

One project can hold several **scenes**, each with its own layers, camera, light and post effects, and play them one after another on a **timeline**. Open  **Scenes & timeline** on the left and press **Start a timeline**: your scene becomes the first clip.

- **Scenes:** + **Empty scene** or + **Copy of this scene**, then ⇌ **edit** to work on one; every panel (layers, camera, light, effects, nodes) now edits that scene. Each scene has its **Own loop**, which keeps looping inside its clips.
- **Clips:** pick the scene and the length in beats, reorder with ↑ ↓. Each clip comes in with a transition over its first beats: **Cut**, **Crossfade**, **Wipe** (with an angle), **Iris**, **Flash** (to white and back), **Glitch** (blocks of the new scene pop in) or **Cut on kick** (with a song: the cut lands on the first kick).
- The whole timeline is the loop (the Timing panel shows its length), so exports play every clip and loop seamlessly: the first clip's transition comes in from the last one.
- Above the preview, **This scene** shows only the scene you are editing and  **Timeline** plays everything. The move/rotate handles only show for the scene being edited.
- With a song loaded, 🎵 **Clips from the song's sections** finds where the music changes (drops, breakdowns, new parts) and makes one clip per part, taking turns through your scenes; the loop becomes the whole song from its start.
- **Stop the timeline** goes back to one looping scene; the other scenes are kept.

The **Scene Tour** preset chains three scenes with glitch, wipe and iris transitions.

# 11 Randomize, Surprise me & undo

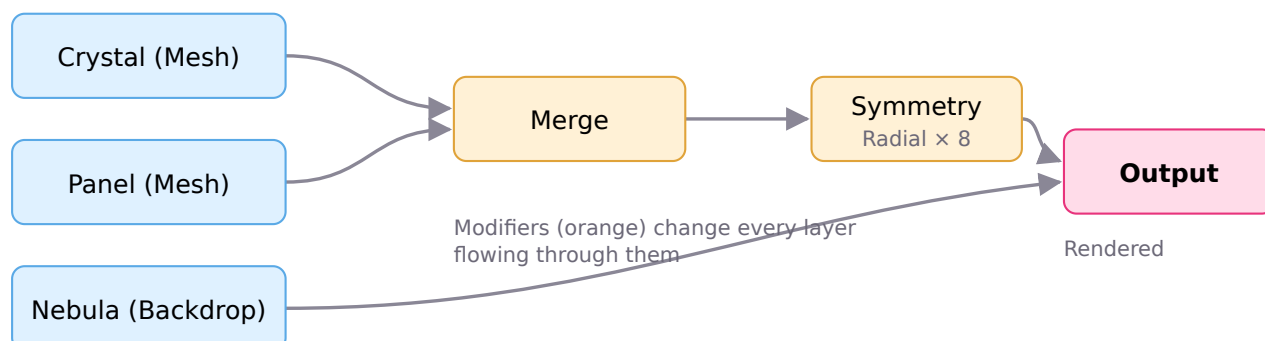
---

- 🎲 **Randomize** shuffles the current scene: one harmonious hue rotation of all colours, new shapes and layouts, new motion and a few post-effect changes. Every click gives a new variation.
- ⚙️ opens the **Randomizer** settings. Choose what may change (Colours, Shapes & layouts, Motion, Post effects), how strongly, and the **seed**. The same seed always gives the same result.
- **Surprise me** picks a random preset and randomizes it strongly.
- **Undo / redo** (`Ctrl + Z` / `Ctrl + Shift + Z`) covers everything, so randomize freely.

## 12 Nodes mode (advanced)

Click **Nodes** in the top bar to build the scene as a **graph**. Layers become **source nodes** that flow through **modifiers** into an **Output**. Everything connected to Output is rendered.

Sources (blue) emit layers

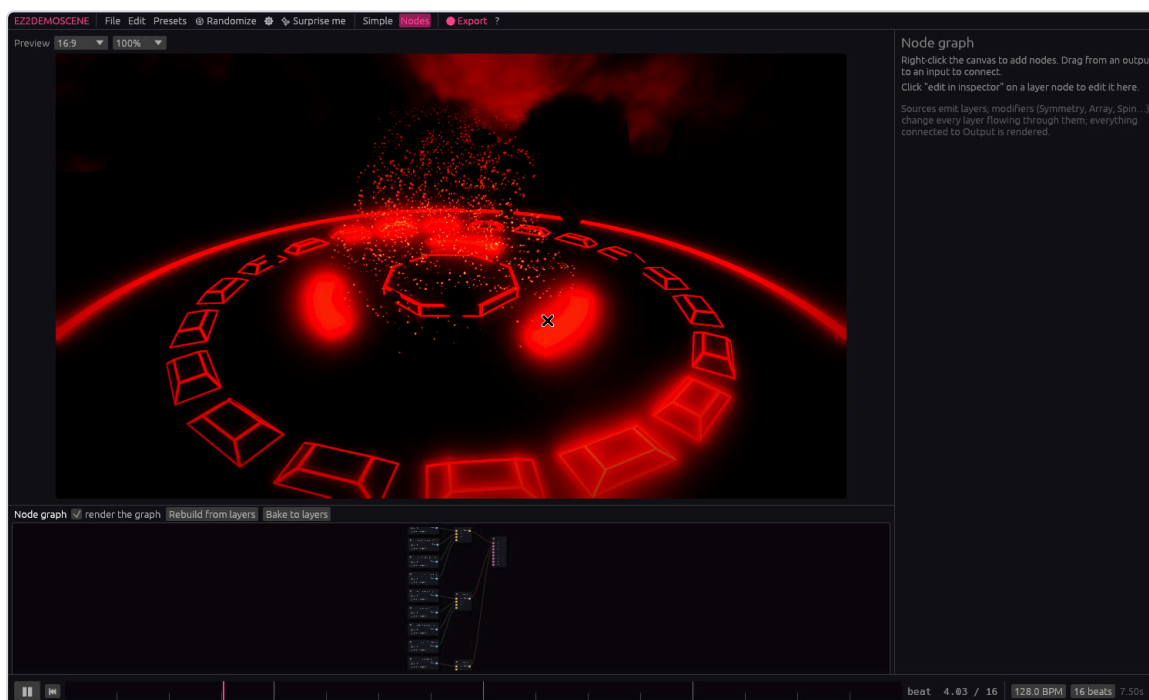


The crystal and panel layers are merged, turned into 8 radial copies, and sent to Output together with the nebula.

| Node                  | What it does                                                                                                                                                                                                                                                                                                    |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source                | One layer. Click <b>edit in inspector</b> to edit it on the right with the usual controls.                                                                                                                                                                                                                      |
| Symmetry              | Replaces the symmetry of every incoming layer.                                                                                                                                                                                                                                                                  |
| Offset                | Moves every incoming layer.                                                                                                                                                                                                                                                                                     |
| Spin                  | Adds turns per loop.                                                                                                                                                                                                                                                                                            |
| Scale                 | Multiplies the size.                                                                                                                                                                                                                                                                                            |
| Tint                  | Rotates hues and scales the glow.                                                                                                                                                                                                                                                                               |
| Array                 | Repeats the incoming layers N times, stepping the position and rotation for each copy.                                                                                                                                                                                                                          |
| Jitter                | Moves and turns every incoming layer by a random amount ( <b>seed</b> reshuffles). With <b>re-roll</b> at 0 it scatters the layers once; above 0 it becomes a beat-synced <b>shake</b> that picks a new direction that many times per loop. <b>Move</b> and <b>Turn</b> have <b>~</b> , so fades work here too. |
| Mirror                | Adds a mirrored copy of every incoming layer, across the X, Y or Z plane at a chosen position.                                                                                                                                                                                                                  |
| Strobe                | Sets the Blink / strobe of every incoming layer (chapter 7).                                                                                                                                                                                                                                                    |
| Colour / material     | Sets the glow colour, multiplies the glow, and for shapes can set a texture, a neon wireframe look and a glitch.                                                                                                                                                                                                |
| Deform                | Sets the Deform of every incoming shape.                                                                                                                                                                                                                                                                        |
| Colours across copies | Sets the colours across the copies of every incoming shape.                                                                                                                                                                                                                                                     |

|               |                                                                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Along a curve | Lays the copies along a curve. Wire a <b>ribbon</b> layer into its second input ( <b>ribbon</b> ) and they follow that ribbon exactly, wherever it is. |
| On a surface  | Scatters the copies over the shape layer wired into <b>surface</b> , following its placement.                                                          |
| On a terrain  | Stands the copies on the terrain wired into <b>terrain</b> .                                                                                           |
| Drive         | Sets one setting of every incoming layer from a signal (see <i>Signals</i> below).                                                                     |
| Merge         | Combines up to 4 streams.                                                                                                                              |
| Output        | Up to 8 inputs. Everything here is drawn.                                                                                                              |

**Working with the graph:** right-click empty space to add nodes, drag from an output dot to an input dot to connect, and right-click a node to **Duplicate** or **Delete** it. The toolbar has three controls: **render the graph** switches between the graph and the plain layer list; **Rebuild from layers** starts a new graph from your layers; and **Bake to layers** turns the graph's result into ordinary layers and returns to Simple mode.



Nodes mode: the viewport on top, the graph below, the inspector on the right.

## Signals: wire any setting to a wave, a beat or the music

Besides layers, wires can carry **signals**: one number that changes over the loop. Signal pins are **green** (outputs are squares). A signal node shows a small graph of its value over the whole loop, with the current moment marked in pink. Add them from the right-click menu under **Signal**.

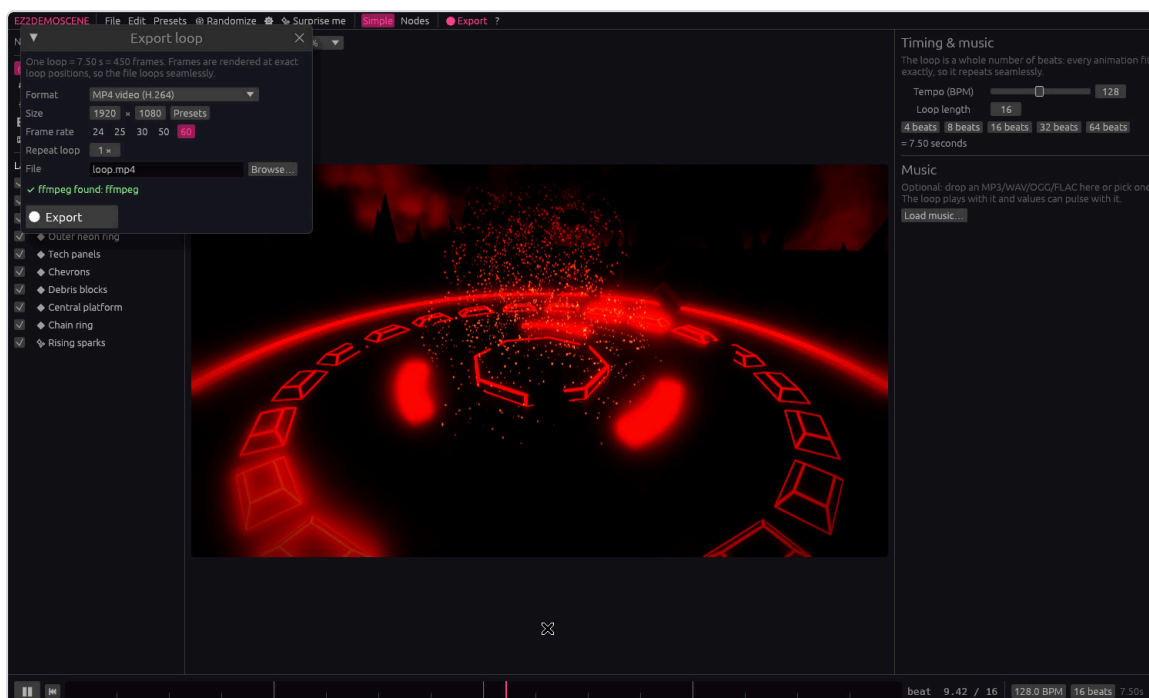
| Signal node  | What it does                                                                                                                                                                                                              |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wave / music | Any animation: a wave with whole cycles per loop, a fade on every beat, random steps, or a link to the music (follow the kick, bass or melody, or react to hits). It uses the same ~ and 🎵 controls as every other value. |

|             |                                                                                                                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Math        | $A + B$ , $A - B$ , $A \times B$ , $A \div B$ , the smaller or larger one, or $A > B$ / $A < B$ (1 or 0).<br>Unconnected inputs use the numbers typed in the node.                                                           |
| Remap       | Maps one range to another, e.g. $-1 \dots 1$ to $0.7 \dots 1.35$ . <b>stay in range</b> clamps.                                                                                                                              |
| Quantize    | Rounds to steps (4 steps per $1 \rightarrow 0, 0.25, 0.5 \dots$ ).                                                                                                                                                           |
| Smooth      | Averages over a few beats around each moment. Great on random steps. Music links have their own smoothing (the 🎵 row), which works better for them.                                                                          |
| Mix         | Blends A to B by T.                                                                                                                                                                                                          |
| Sequence    | Steps through your list of values, one every N beats, starting over with the loop. <b>glide</b> slides from each value to the next.                                                                                          |
| Hit counter | Counts kicks (or snares, hats, any hit, notes) since the start of the loop: 0, 1, 2... up to <b>count to</b> – 1, then starts again. Through a Remap ( $0 \dots 3 \rightarrow 0 \dots 1$ ) it steps a setting on every kick. |

A **Drive** node puts a signal to work. Wire layers into its first input and a signal into the second, then pick the **setting** from the list: `kind. ....` are the layer's own settings (for a shape, `kind.material.emissive` is its glow), `transform. ....` its placement, and `.0 .1 .2` are x, y, z (or red, green, blue). Choose how the signal is used: **set to** replaces the value, **add** adds to it (the setting keeps its own animation), and **multiply by** scales it. Chain Drive nodes to drive several settings.

Signals are worked out from the loop position and the music at that moment, so graphs always loop seamlessly. The **Signal Flow** preset shows a stepped glow Sequence and a smoothed random size on the centrepiece. **Bake to layers** drops the Drive nodes (settings keep their own values).

# 13 Exporting your loop



The export dialog ( **Ctrl** + **E** ).

| Option        | Notes                                                                                                                                                                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Format        | <b>MP4 (H.264)</b> works everywhere. <b>WebM (VP9)</b> is for the web. <b>Animated GIF</b> is for chats and forums. <b>PNG sequence</b> writes one PNG per frame into a folder and needs no ffmpeg; it is ideal for video editors and VJ software.                                        |
| Size          | Type a size or use <b>Presets</b> : 720p, 1080p, 1440p, 4K, Square, Vertical (phone), GIF small, Retro 4:3.                                                                                                                                                                               |
| Frame rate    | 24, 25, 30, 50 or 60 fps. The dialog shows how many frames one loop needs.                                                                                                                                                                                                                |
| Motion blur   | Averages 4, 8 or 16 in-between moments per frame for smooth, film-like motion (fast spins and flying debris smear instead of stepping). <b>Shutter</b> sets how much of the time between frames is blurred. The export takes that many times longer, and the video still loops perfectly. |
| Repeat loop   | Makes the file longer by repeating the loop. Useful for platforms that don't loop videos.                                                                                                                                                                                                 |
| Music         | Include the music track (MP4/WebM only).                                                                                                                                                                                                                                                  |
| File / Folder | Where to save. Use <b>Browse....</b>                                                                                                                                                                                                                                                      |

Press **Export**. A progress bar shows the frames being rendered, and you can keep working or press **Cancel**. Export renders every frame at full quality (anti-aliasing and full-resolution effects) even if your preview is set lower.

### GIF tips

GIFs get big quickly. Use 480×270 at 25–30 fps, keep loops short (8 beats), and avoid film grain, which makes GIFs much larger.

### "ffmpeg not found"?

Install ffmpeg, then either make sure `ffmpeg` runs in a terminal or paste the path to the `ffmpeg` program into the box in the export dialog and press **Check**.

## Stills

**File → Save still image...** saves the current frame as a PNG at the export size.

## Command line

You can render without opening the window, which is handy for batch jobs:

```
ez2demoscene --list-presets
ez2demoscene --export friend_scene.ez2pack loop.mp4 # packs work too
ez2demoscene --render "Neon Arena" still.png --phase 0.25 --size 1920x1080
ez2demoscene --export "Gold Kaleido Room" loop.mp4 --size 1920x1080 --fps 60 --repeats 4
ez2demoscene --export "Orbiting Solid" smooth.mp4 --fps 30 --motion-blur 8 --shutter 0.5
ez2demoscene --export my.ez2.json frames/ # PNG sequence into a folder
ez2demoscene --write-presets my_presets/ # presets as editable project files
ez2demoscene --write-textures my_textures/ # the retro texture pack as PNGs
```

`--phase` is a position in the loop from 0 to 1 (0.5 is the middle).

# 14 Saving, sharing & your library

---

## Project files

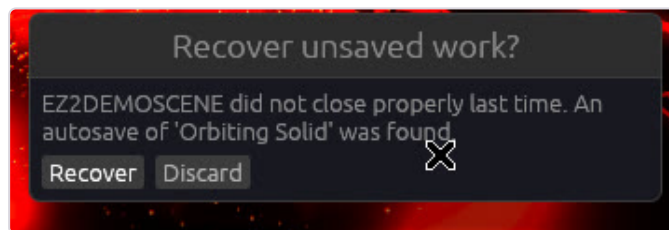
**File → Save** (**Ctrl** + **S**) writes a `.ez2.json` project. Models, images and music stored **inside the project's folder** (or its sub-folders) are remembered **relative to the project**. You can move or copy the whole folder, even to another computer, and everything is still found. Files elsewhere on your disk are remembered by their full path.

## Packs: one file with everything

**File → Save as pack (.ez2pack)...** writes a single file containing the project **and copies of every model, image and music file it uses**. It's the best way to send a scene to a friend or archive it. Open a pack with **File → Open...** or by dropping it on the window. It is unpacked into your library folder, and **Save** then asks where to keep the project. If some files were missing when packing, the status bar lists them.

## Autosave & crash recovery

While you work, unsaved changes are autosaved every 30 seconds. If the program or computer crashes, the next start offers to **recover** them:

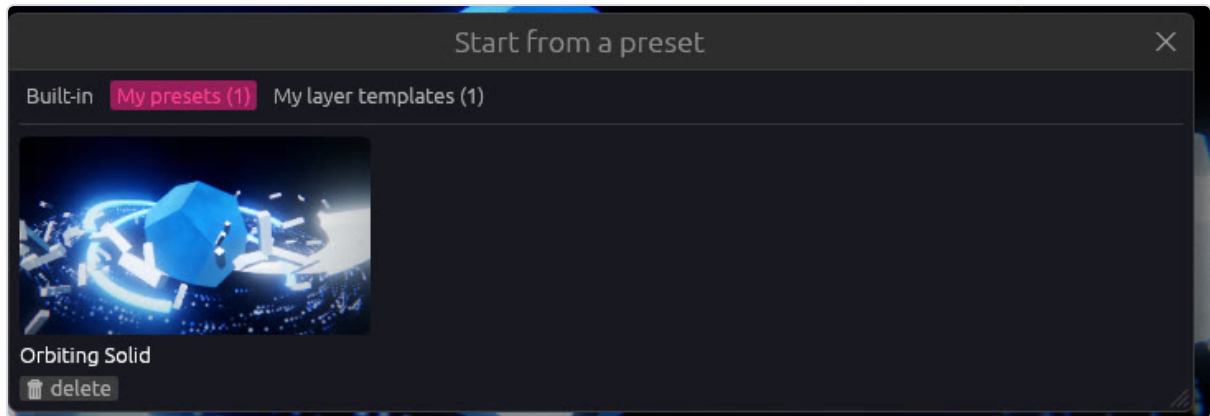


Shown after a crash. **Recover** brings back the scene; **Discard** throws the autosave away.

After recovering, save the project somewhere with **Ctrl** + **S**. When you quit normally, the autosave is deleted.

## My presets

**File → Save as my preset...** adds the current scene to the gallery with a thumbnail of the current frame. Find it under **Presets → My presets**. Clicking one opens it as a new untitled project, so your preset is never overwritten by accident. The  **delete** button removes it.



The gallery's **My presets** tab.

## Layer templates

Built a great particle effect or neon ring? **Right-click the layer → Save as template**. It then appears in **+ Add → ★ My templates** in every project, and also in the node graph's right-click menu. Manage your templates in **Presets → My layer templates**.

### Where is my library?

Your presets, templates, unpacked packs and autosave live in your user data folder:

`%APPDATA%\ez2demoscene\EZ2DEMOSCENE\data` on Windows, `~/.local/share/ez2demoscene` on Linux, and `~/Library/Application Support/org.ez2demoscene.EZ2DEMOSCENE` on macOS. Copy it to back everything up.

# 15 Phones, tablets & the web

EZ2DEMOSCENE also runs **in a web browser**, with nothing to install. It's the same editor with the same presets, layers, node graph and exports. Open `https://roganis.github.io/ez2demoscene/` in Chrome, Edge or Firefox on a computer, or in Chrome on an Android phone or tablet. On Android you can also install the **EZ2DEMOSCENE app (APK)**, which bundles the same editor and works offline.

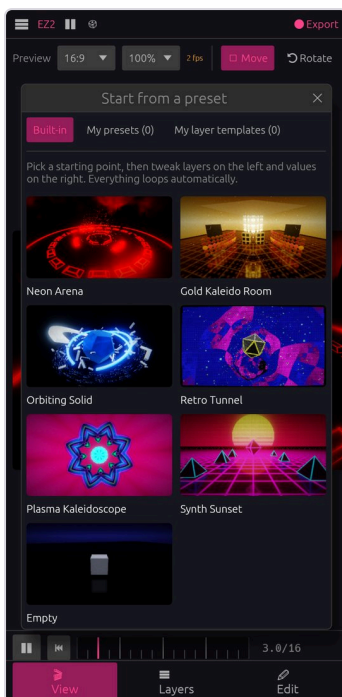
## Installing

| How                   | Steps                                                                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Browser               | Just open the link. The first visit downloads about 5 MB; after that it loads from the cache and also works offline.                                                                                  |
| Home-screen app (PWA) | In Chrome, open the <b>⋮</b> menu → <b>Add to Home screen</b> / <b>Install app</b> . It opens full-screen like a normal app.                                                                          |
| Android APK           | Download <code>app-debug.apk</code> (or the signed release APK) from the project's GitHub Releases page, open it and allow "install unknown apps" for your browser or file manager when Android asks. |

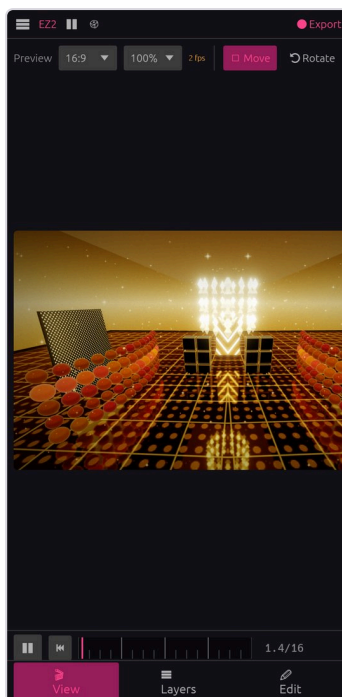
## The phone layout

When the window is narrower than about 820 pixels, as on a phone held upright, the editor switches to a **one-panel layout**. The top bar holds **≡** (all menus), play/pause, **🎲** (Surprise me) and **Export**. A compact timeline sits above a **tab bar** with three tabs:

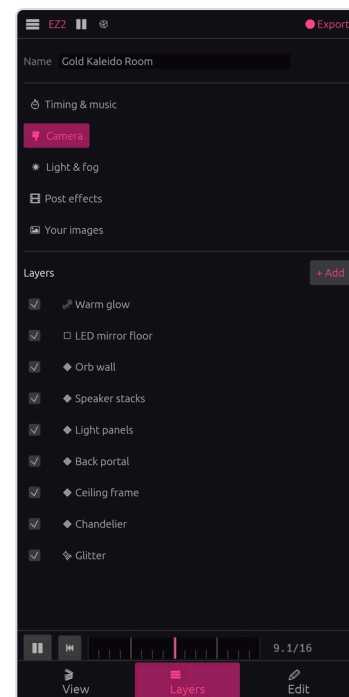
- **View**: the live preview, with the Move/Rotate gizmos.
- **Layers**: the scene settings and layer list (or the **Graph** in Nodes mode).
- **Edit**: the inspector for whatever you selected. Tapping a layer, a scene item or a node jumps here automatically.



Preset gallery on a phone.



The **View** tab.



The **Layers** tab.

Turn a tablet sideways (or use a desktop browser) to get the full desktop layout with all panels side by side.

## Touch gestures in the preview

| Gesture                      | What it does                                                                                 |
|------------------------------|----------------------------------------------------------------------------------------------|
| One finger on a gizmo handle | Moves or rotates the selected layer, as with the mouse. Handles are bigger on touch screens. |
| One finger elsewhere         | Orbits the camera around the scene.                                                          |
| Two-finger pinch             | Moves the camera closer or further away.                                                     |
| Two-finger twist             | Turns the camera around the scene.                                                           |
| Two fingers up/down          | Raises or lowers the camera.                                                                 |
| Long press on a layer        | Opens its menu (duplicate, delete, save as template...), like a right-click.                 |

## Files in the browser

- **Importing** models, images and music opens your device's file picker; you can also drop files onto the page on a computer. Imported files are kept inside the browser, so they're still there next time.
- **Saving:** **File** → **Save** downloads a **pack** ( `.ez2pack` ) with the scene and all its files. Open it again with **File** → **Open...**, on the web or in the desktop app. In the Android app, saving and exporting open Android's **Save as** dialog: pick a folder (Downloads, Documents, Drive...) and tap **Save**. From there you can share the file to a chat with the Files app.
- **My presets, layer templates and the autosave** live in the browser's storage (IndexedDB). If you come back with unsaved changes, the editor offers to **recover** them. Clearing the browser's

site data deletes this library, so save packs of anything you want to keep.

## Exporting on the web

| Format            | Notes                                                                                                                                                                                                                                                                                |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MP4 / WebM video  | Encoded by the browser's own video encoder (WebCodecs, hardware-accelerated on most phones). Loops stay frame-exact. MP4 uses H.264 where available, otherwise VP9 or AV1. <b>The web version exports video without the music track</b> ; use the desktop app for videos with sound. |
| GIF               | Works everywhere; loops forever by itself. Keep it small (e.g. 480×270) for sharing.                                                                                                                                                                                                 |
| PNG frames (.zip) | Every frame as a lossless PNG, for editing elsewhere.                                                                                                                                                                                                                                |

The preview pauses while exporting so the device can concentrate on the frames. Phones are slower than desktop PCs, so start with 720p or less, and keep the tab open until the download starts.

### WebGPU and WebGL

On computers, recent Chrome and Edge render with **WebGPU**, the fastest path. Other browsers, and all Android phones and tablets, use **WebGL2**, which looks the same (some phone GPUs corrupt the picture with WebGPU). If a scene stutters, lower **Preview** to 50% in the View tab; exports are unaffected.

## Graphics settings

☰ → 🖼️ **Graphics** (the 🖼️ button on a computer) shows which backend, GPU and driver the app is using; 📋 **Copy** copies the details for a bug report. On the web and in the Android app you can also force **WebGL2** or **WebGPU** there. The change applies after ↻ **Restart now**. If the chosen backend can't start, the app goes back to **Automatic** by itself.

# 16 Recipes

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## Neon arena (black gloss and red neon)

1. Background **Nebula clouds**, with very dark colours A/B and a red C. Set the fog colour to almost-black red.
2. Add a **Mirror floor**: colour black, Reflection 0.35, Blur 0.3.
3. Add **Crystal cluster** with Copies → Radial (count 50, radius 34), Size 7, Random tilt 18, Random size 0.5. Use a glossy black material (metallic 0.9, roughness 0.15, Faceted).
4. Add **Panel** with Radial (20, radius 11) and Stretch 2.6 / 0.35 / 1.2. Glow red, **Edges**, strength 5, and Light chase 1.
5. Add **Ring band** (Width 0.02) stretched 16 / 1 / 16 with red glow, Full, and a Sine animation.
6. Add **Vortex** particles in orange → red. Turn on Bloom 1.1 and Chromatic aberration.

## Golden mirrored room

1. Background **Gradient sky** in warm golds. Set the light to warm white and the ambient to 0.45.
2. Add a **Mirror floor** with Reflection 0.85, Reflection tint gold, texture `led_grid` and Grid glow 0.6.
3. Add **Sphere** with Copies → Wall (14 × 5, curve 50°), Size 0.33, Rotation y 90, Position x 6.5, Symmetry **Mirror X**, and Light chase 0.6.
4. Add **Cube** with texture `speaker`, a Grid of 2 × 3 and Symmetry Mirror X. Pulse the glow on the beat.
5. Add **Octahedron** with Spiral copies high above the floor as a chandelier, plus **Snow / glitter** star particles.
6. Post: **Mirror split** left → right and Bloom 1.0.

## Crystal orbited by debris

1. Background **Starfield**.
2. Add **Dodecahedron**, Size 1.6, blue, metallic 0.5, roughness 0.18, Faceted, Rim light 0.9, Spin / loop x 1, y 1.
3. Add **Cube** with Copies → Orbit swarm (70, radius 3.2, spread 0.9), Stretch 0.12 / 0.12 / 0.5 and Random size 0.6.
4. Add **Ring band** arcs (Arc 250°) with blue glow and Spin / loop y 2, plus **Ring orbit** particles with trails.

## 90s demo tunnel

1. Background **Raymarched tunnel** with Wall texture `xor` and Motion / loop 4.
2. Add **Icosahedron** with texture `checker`, Chunky pixels, Glow Edges, and Spin / loop 2 / 1 / 0.
3. Add **Warp stars** with the Pixel square sprite and Trail 4.
4. Post: Pixelate 3, Retro palette EGA with dithering 0.7, CRT monitor, and Beat flash 0.15.

## Stormy lake

1. Background **Volumetric clouds**, variant **Storm**, with grey-blue colours. Fog colour dark grey-blue, Sun strength 0.7.
2. Add **Terrain**: Solid, Shape **Ridged mountains**, Biome **Alpine**, Valley 0.45, and Liquid **Water** at Level 0.12.
3. Add **Weather** → **Rain** with Falls / loop 14 and a Sine on Wind for gusts. Tick **Lightning**.
4. Post: lower Saturation to 0.8 and add a little Film grain.

## Lava world

1. Background **Volumetric clouds** (Overcast) in dark reds, with the sun low in front of the camera.
2. Add **Terrain**: Shape **Canyons**, Biome **Volcanic**, Liquid **Lava** at Level 0.3, Current / loop 2. Give the lava Glow a slow Sine.
3. Add **Weather** → **Rising embers** with a strong Brightness so they bloom.
4. Post: **God rays** with a warm tint, and Bloom.

## Sunken temple (underwater)

1. Background **Gradient sky** from light blue (top) to deep blue. Fog colour deep blue, Fog density 0.04.
2. Turn on **Underwater caustics** (Amount 1.6, pale blue).
3. Add a **Mirror floor** with Reflection 0.05 and a sandy colour, pillars (Cylinder, Radial copies) and a glowing gem.
4. Add **Fountain** particles with the Ring sprite as bubbles, and **God rays** with a low threshold.

# 17 Reference

## Keyboard & mouse

| Action                       | Shortcut                                                  |
|------------------------------|-----------------------------------------------------------|
| Play / pause                 | Space                                                     |
| Save                         | Ctrl + S                                                  |
| Open                         | Ctrl + O                                                  |
| Export                       | Ctrl + E                                                  |
| Undo / Redo                  | Ctrl + Z / Ctrl + Shift + Z (or Ctrl + Y)                 |
| Move / Rotate / Scale tool   | W / E / R                                                 |
| Ground grid on/off           | G                                                         |
| Select an object             | Click it in the viewport                                  |
| Snap while dragging a handle | Hold Ctrl                                                 |
| Turn the camera              | Drag in the viewport (not on a handle)                    |
| Zoom the camera              | Mouse wheel over the viewport                             |
| Scrub time                   | Click or drag in the timeline                             |
| Import                       | Drop project, model, image or music files onto the window |

## Performance

The small **fps** number above the viewport shows how smoothly the preview runs. It turns orange below 24 fps. **Hover it** to see the frame time, triangle and particle counts, whether the mirror floor is drawing the scene a second time, and the **five most expensive layers**. A  next to a layer in the list marks a heavy layer. Hover it for details. The usual fixes are fewer copies, less sphere detail, fewer particles or a shorter trail. You can also lower the preview quality, which only affects the preview and never your exports. Volumetric clouds and the raymarched backgrounds are the heaviest layers; their **Quality** setting trades detail for speed.

## Troubleshooting

| Problem                                 | Solution                                                                                                                                                                                            |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Playback stutters                       | Hover the fps readout to find the heaviest layers (see Performance above), or lower the preview quality. The fractal background and big particle counts are the heaviest. Exports are not affected. |
| The computer crashed and I lost my work | Start EZ2DEMOSCENE again and choose <b>Recover</b> (chapter 14).                                                                                                                                    |

|                                                             |                                                                                                                                                                                       |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A friend's project can't find its files                     | Ask them to send a <b>pack</b> ( <b>File → Save as pack</b> ) instead of the <code>.ez2.json</code> file.                                                                             |
| Everything is washed out or one colour                      | A big glowing object is probably filling the screen. Reduce Glow, raise the Bloom threshold, or check the Size/Stretch of ring and panel layers.                                      |
| A shape is invisible                                        | Check its visibility box, its Size, whether it is below the floor or inside another object, and the fog density.                                                                      |
| My model shows as a cube                                    | The file couldn't be read. The reason appears in red at the bottom of the inspector. Use <b>Edit → Reload models &amp; images</b> after fixing the file.                              |
| No sound                                                    | No audio output device was found. The music is still used for sync, ♪ modulation and export.                                                                                          |
| Video export fails                                          | Check ffmpeg (chapter 13), or export a PNG sequence instead.                                                                                                                          |
| The picture corrupts or flickers on a phone or in a browser | Open  <b>Graphics</b> and switch to the other backend (WebGL2 or WebGPU), then restart (chapter 15). |
| The app doesn't start                                       | Update your graphics drivers. On Linux, install Vulkan drivers (e.g. <code>mesa-vulkan-drivers</code> ).                                                                              |

## Project files

Projects are saved as readable `.ez2.json` text files. Imported models, images and music are stored as **file paths**, so keep the files inside the project's folder (they're then stored relative to it), or share a **pack** (chapter 14). The built-in presets are also available as project files in `assets/presets`.

## Glossary

| Term            | Meaning                                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------------------|
| Beat / BPM      | A musical pulse. BPM is beats per minute.                                                                          |
| Loop / phase    | The repeating clip. Phase is a position in it, from 0 (start) to 1 (end, which equals the start).                  |
| Instancing      | Drawing many copies of one shape efficiently.                                                                      |
| Emission / glow | Light a surface gives off by itself.                                                                               |
| Bloom           | The halo that bright things spill onto their surroundings.                                                         |
| Raymarching     | A technique that draws shapes defined by maths (tunnels, fractals) instead of polygons. It is a demoscene classic. |
| Dithering       | Patterns of pixels that fake more colours than a palette has.                                                      |
| Demoscene       | A computer art subculture, dating back to the 1980s, that makes real-time audiovisual "demos".                     |